



OPEN Dynamic evolution of intracortical and corticomuscular connectivity during reach-and-grasp movement planning and execution

Jorge A. Sanchez-Bautista, Javier M. Antelis & Omar Mendoza-Montoya

Reach and grasp movements involve a complex transformation of information from visual to motor space. Initially, task-related information is encoded in extrinsic, object-centered spatial coordinates, whereas motor output is represented in intrinsic, body-centered coordinates. In this study, effective connectivity was assessed using Partial Directed Coherence (PDC) to examine both intracortical and corticomuscular interactions during self-initiated reach and grasp actions. EEG signals from 16 cortical channels and EMG signals from 6 upper limb muscles were analyzed in 20 healthy right-handed participants (aged 18 to 45) performing vertical and lateral grasping tasks with a self-selected limb. Intracortical connectivity was evaluated across four distinct time windows spanning the planning and execution phases, while corticomuscular connectivity, defined as directed interactions from EEG to EMG, was assessed continuously from one second before to one second after movement onset. The results revealed that intracortical connectivity evolved dynamically across the premovement and execution phases, reflecting a continuous modulation of cortical interactions during motor planning and performance. Corticomuscular connectivity from EEG to EMG consistently decreased around movement onset across all evaluated conditions. Although some differences were observed between the PDC curves related to the grasp types and the used limb, they did not reach statistical significance after correction for multiple comparisons. These findings highlight the temporal specificity and task dependence of motor-related connectivity, offering insights into the distributed neural processes underlying sensorimotor transformations.

Keywords Reach-and-grasp movement, Functional connectivity, Partial directed coherence (PDC), Electroencephalography (EEG), Electromyography (EMG)

Voluntary movement arises from hierarchical sensorimotor transformations that convert sensory inputs into appropriate motor outputs^{1,2}. In this process, extrinsic information (e.g., target location) and intrinsic signals (e.g., joint angles or muscle forces) are integrated to estimate the current state of the body and environment^{1,3,4}. Based on these internal representations, motor commands are generated through both feedforward and feedback mechanisms. Feedforward control supports rapid initiation through anticipatory motor outputs, while feedback control refines ongoing movement based on sensory error signals^{1,3,5–8}.

Reach-and-grasp movements are among the most extensively studied forms of self-initiated actions due to their critical role in interacting with objects and the environment^{9,10}. These actions involve a transformation from visual to motor space, in which task-related information is initially encoded in extrinsic coordinates centered on objects and later converted into intrinsic coordinates centered on the body to generate motor output^{11,12}. Neural activity associated with these transformations has been observed across multiple cortical regions, particularly in the primary motor cortex (M1), where preparatory signals can emerge up to 0.9 seconds before movement onset^{13–17}.

A key aspect in understanding cortical dynamics during movement planning and execution is connectivity, defined as the temporal relationship between spatially distributed neurophysiological signals. This concept includes structural, functional, and effective connectivity^{18–21}. Structural connectivity refers to the anatomical pathways that link different regions of the brain. Functional connectivity captures statistical dependencies

Escuela de Ingeniería y Ciencias, Tecnológico de Monterrey, Monterrey, NL 64849, Mexico. email: omendoza83@tec.mx

between neural signals. In contrast, effective connectivity estimates the directed influence that one region exerts over another, based on generative models of neural interactions and task-related timing²².

Several previous studies on functional connectivity have focused on the execution phase of movement, reporting increased connectivity in the alpha and beta frequency bands compared to baseline conditions^{23–26}. With regard to the premovement phase, Kim et al²⁷ analyzed connectivity during the Bereitschaftspotential and reported distinct connectivity patterns in the beta and gamma bands.

Among the available metrics for effective connectivity, Partial Directed Coherence (PDC) stands out for its ability to characterize directed interactions in the frequency domain using multivariate autoregressive (MVAR) modeling. Derived from the framework of Granger causality, PDC enables the identification of both direct and indirect influences between brain regions, as well as between cortical and muscular signals^{28–30}. Previous studies have applied PDC to EEG or EEG-EMG data to explore motor control under various task conditions^{19,20}. Table 1 highlights a selection of such studies, showing the range of motor paradigms explored and the increasing interest in tracking how connectivity changes over time.

As shown in Table 1, previous research has used PDC and variants (e.g., E-PDC) to investigate effective connectivity during a wide range of motor tasks, including postural control, object manipulation, and simple or externally cued limb movements. These studies have provided valuable insights into task-dependent cortical and corticomuscular interaction patterns. However, most have examined conditions with fixed timing or externally imposed constraints, and only a few have addressed how connectivity evolves during self-paced actions initiated at the participant's discretion. In particular, there is limited evidence on how voluntary decisions regarding when and with which limb to initiate a movement modulate directed connectivity across planning and execution phases.

Building upon our previous work¹⁷, which demonstrated that relevant information related to upcoming self-initiated reach-and-grasp movements can be extracted during the premovement phase, the present study extends this approach by characterizing intracortical and corticomuscular connectivity across distinct planning and execution time windows. Importantly, prior studies have shown that when participants are instructed to initiate a movement immediately after an external cue (e.g., a “go” signal or visual stimulus), the resulting brain activity is likely to be influenced by that stimulus rather than reflecting purely voluntary processes³⁸. This issue is particularly critical for research aiming to detect movement intention, especially in the context of Brain–Computer Interfaces (BCIs) for neurorehabilitation, where self-selected and self-initiated movements by the user are considered essential³⁹. In fact, several investigators have instructed participants to delay their response after a cue to minimize evoked responses in the premovement activity^{39,40}.

Regarding corticomuscular coherence (CMC), it has traditionally been examined during periods of sustained muscle activation, reflecting the synchronization between cortical and spinal muscular activity during movement execution. However, some studies investigating isometric motor tasks have reported weak but significant corticomuscular coupling even before movement onset, suggesting that descending cortical drive may already be functionally engaged during motor preparation^{41–44}. These findings indicate that CMC may not only reflect ongoing sensorimotor integration but also preparatory communication between cortical and peripheral structures.

Although a few studies have attempted to evaluate CMC during dynamic or phasic movements, their results have generally shown weaker and more transient coherence compared to isometric contractions, likely due to the non-stationary nature of muscle activation in such tasks^{45–47}. This limitation has constrained our understanding of how cortical activity dynamically interacts with peripheral motor output during natural movements.

To address these gaps in understanding voluntary movement and the limitations of current acquisition protocols, the present study examines the temporal evolution of intracortical and corticomuscular connectivity during self-initiated reach-and-grasp movements. Participants freely selected both the timing of movement onset and the limb used to perform each action, allowing for a natural expression of voluntary motor control

Study	Task	Signals	Temporal Resolution	Dynamic PDC	Main contribution
Astolfi et al. 2006 ³¹	Self-paced foot movements.	EEG	1.5 s	No	Assessment of cortical functional connectivity using PDC/sPDC on real EEG data.
Boonstra et al. 2015 ³²	Postural control while standing upright.	EMG	60 s	No	Muscle connectivity analysis with EMG during postural control.
Tropini et al. 2011 ³³	Visually guided choice reaction task using a joystick.	EEG	480 ms	Yes	Detection of altered directional connectivity in Parkinson's disease with EEG through the PDC.
Eldeeb et al. 2019 ³⁴	Grasping and releasing functional objects.	EEG	400 ms	Yes	Introduction of E-PDC for analyzing post-stroke connectivity using EEG.
Peterson et al. 2019 ³⁵	Postural control with perturbations.	EEG + EMG	400 ms	Yes	corticomuscular connectivity in postural control with disturbances.
Kenville et al. 2020 ³⁶	Bipedal squat (BpS)	EEG + EMG	Not reported	Yes	Dynamic corticomuscular control during squatting.
Hua et al. 2024 ³⁷	Postural control with sensory conflicts.	EEG	5 s	Yes	Cortical network reconfiguration in response to sensory conflicts.

Table 1. Summary of studies using Partial Directed Coherence (PDC) and variants (e.g., E-PDC) to analyze effective connectivity during motor tasks. The reviewed works span simple and complex movements, including postural control and whole-body actions. Recent studies increasingly apply dynamic PDC to capture time-varying connectivity. Main contributions include evaluating motor-related connectivity, identifying clinical alterations, and advancing PDC methodologies.

while minimizing evoked responses. EEG and EMG signals were analyzed using time-resolved Partial Directed Coherence, enabling the characterization of directed connectivity across distinct planning and execution windows. This approach reveals how connectivity patterns change over time depending on task demands and effector choice, offering new insights into the neural dynamics underlying goal-directed movement.

Results

EMG-based movement onset, duration, and retained trials

Movement onset was identified across all trials based on EMG signal power. The moment of cue presentation, that is, the visual stimulus indicating that participants were allowed to initiate movement, was used as the temporal reference point ($t = 0$) to determine the latency of movement initiation. Fewer than 3% of trials showed movement initiation within 1 second of cue onset, and all movement onsets occurred within 7 seconds. On average, 273 ± 16 trials per participant (range: 257–300) were retained after the trial rejection process, with a mean movement onset time of 3.39 ± 0.98 seconds, indicating consistent responses within the expected temporal window. The duration of each reach-and-grasp movement was estimated using the signals from the cylindrical target device, which was equipped with embedded infrared sensors capable of detecting contact and distinguishing between the different types of grasp executed. Across all participants, the mean movement duration was 0.78 ± 0.17 seconds.

Table 2 summarizes, for each participant, the number of valid trials retained after artifact rejection, as well as their distribution by limb (left/right) and movement type (vertical/lateral). The table also reports descriptive statistics of movement onset latency (mean, standard deviation, maximum, and minimum values, in seconds) relative to the visual signal from movement. Although two participants (IDs 7 and 10) showed a marked imbalance between the number of trials for the dominant and non-dominant limb, the overall distribution was reasonably balanced across conditions for the majority of participants.

To further characterize artifact-related exclusions, we also quantified the number of times each EEG electrode was rejected across participants. Frontal electrodes were most frequently excluded (FP1: 14 instances; FP2: 9; Fz: 8), whereas most central and posterior electrodes exhibited few or no exclusions (typically ≤ 4 instances, with many at 0), indicating that dropout was not systematic. For EMG, some trials were rejected due to transient noise when electrodes became slightly displaced during task execution—usually when participants adjusted their posture at the beginning of an experiment. Importantly, only three participants exhibited this behavior outside the designated rest period, and the affected trials represented a small proportion of the total dataset. No specific muscle channel showed a consistent dropout pattern.

Participant	Number of trials					Movement Onset Time (s)			
	Total	Left Limb	Right Limb	Vertical mov	Lateral Mov	mean	std	max	min
1	278	129	149	145	133	4.69	1.19	6.89	1.40
2	298	155	143	148	150	2.74	0.63	4.73	1.39
3	258	124	134	133	125	3.90	0.84	6.19	1.29
4	259	126	133	131	128	3.90	0.60	6.73	2.49
5	276	137	139	140	136	4.23	0.94	6.69	1.14
6	261	135	126	130	131	3.80	0.92	6.63	1.24
7	281	174	107	139	142	4.25	0.92	6.97	1.06
8	295	156	139	146	149	3.40	0.69	5.68	1.04
9	274	149	125	140	134	2.97	0.60	6.11	1.16
10	265	176	89	135	130	2.17	0.48	3.88	1.11
11	269	136	133	129	140	2.02	0.61	4.14	1.02
12	286	138	148	141	145	1.85	0.36	3.18	1.10
13	273	134	139	131	142	4.24	0.91	6.84	1.76
14	261	119	142	132	129	3.65	0.87	5.94	1.24
15	270	141	129	134	136	2.66	0.88	5.51	1.04
16	275	129	146	135	140	3.10	0.49	4.20	1.79
17	271	139	132	132	139	4.94	0.83	6.94	1.93
18	274	147	127	139	135	4.74	0.88	6.76	1.53
19	257	129	128	124	133	2.47	0.69	4.25	1.18
20	269	132	137	142	127	2.01	0.48	3.36	1.01

Table 2. Number of trials and movement onset times per participant. The table shows, for each participant, the total number of trials and their distribution by limb (left/right) and movement type (vertical/lateral). Descriptive statistics of movement onset times (mean, standard deviation, maximum, and minimum, in seconds) are also provided.

Event-related desynchronization/synchronization analysis

Figure 1 displays the grand average event-related desynchronization/synchronization (ERDS) computed from EEG signals after applying the surface Laplacian transform. These signals correspond to the same set of channels used in the multivariate autoregressive (MVAR) model for the Partial Directed Coherence (PDC) analyses. ERDS was calculated for the preparation and execution phases of all self-initiated reach-and-grasp movements, regardless of the upper limb used or the grasp type. The resulting values were then averaged across all subjects. The abscissa represents time (from 1 to 1 s), and the ordinate represents frequency (from 4 to 40 Hz). Movement onset is indicated by a red vertical line at $t = 0$ s. A white vertical line was added at $t = -0.5$ s to aid interpretation of the timing of synchronization and desynchronization onsets. Desynchronization is shown in blue (negative percentage values), synchronization in yellow (positive percentage values), and non-significant changes in green (zero percentage values). All synchronization and desynchronization patterns correspond to statistically significant changes ($p < 0.05$).

A theta-band (4–8 Hz) synchronization was observed across most electrodes (representing Laplacian-transformed signals), beginning approximately 0.4 seconds before the onset of movement ($t = 0$), fading around 0.2 seconds, and reappearing near $t = 0.7$ s. In the alpha (8–12 Hz), low beta (12–20 Hz), and high beta (20–30 Hz) bands, a sustained desynchronization was detected starting around 0.5 seconds before movement onset and often persisting into the execution phase. Exceptions to this pattern were identified in the Laplacian signals at F3 and F4, where a low beta synchronization appeared around $t = -0.4$ s, and in Fc5, where the synchronization and desynchronization clusters were significantly smaller than in other channels analyzed. In frequencies above 30 Hz, a gamma-band synchronization was observed in nearly all electrodes beginning around $t = -0.3$ s and continuing after movement onset. In Fz, however, this synchronization diminished shortly after movement initiation and re-emerged around $t = 0.8$ s. Overall, theta and gamma band synchronizations tended to lose statistical significance shortly after movement onset, often exhibiting temporal gaps between approximately 0.2 and 0.7 seconds, before reappearing in several channels.

Intracortical connectivity patterns during vertical and lateral grasping

To examine the evolution of cortical connectivity patterns across grasping conditions, partial directed coherence (PDC) was computed between the Laplacian-transformed EEG signals, which enhances local cortical activity by minimizing volume conduction effects. The analysis focused on self-initiated vertical and lateral reach-and-grasp tasks, executed with either the right or left upper limb, as chosen by the participants. Connectivity was evaluated within four time windows aligned to movement onset (0.6 to 0.3 s, 0.3 to 0 s, 0 to 0.3 s, and 0.3 to 0.6 s), and across five EEG frequency bands: theta (4–8 Hz), alpha (8–12 Hz), low beta (12–20 Hz), high beta (20–30 Hz), and gamma (>30 Hz). This approach revealed condition-specific connectivity profiles, highlighting how the frequency and directionality of effective cortical interactions varied with grasp type and the limb engaged.

To visualize these effects, connectivity maps were generated for each frequency band and grasping condition, contrasting executions performed with the right and left upper limbs. Only statistically significant connections

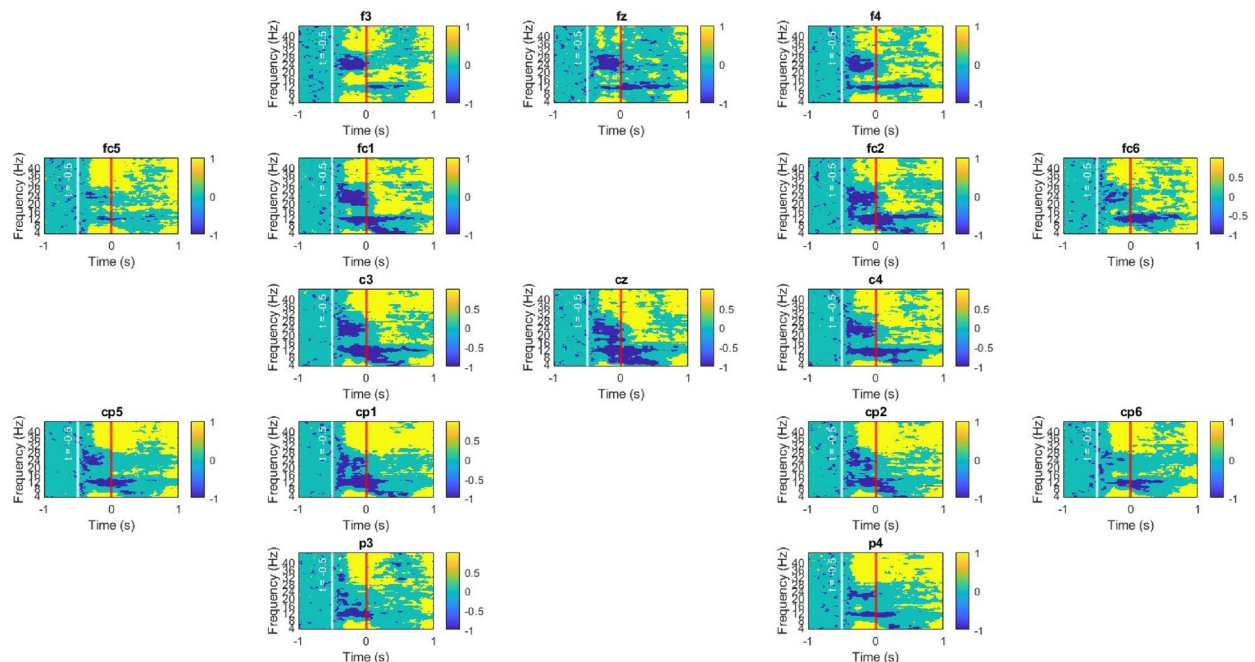


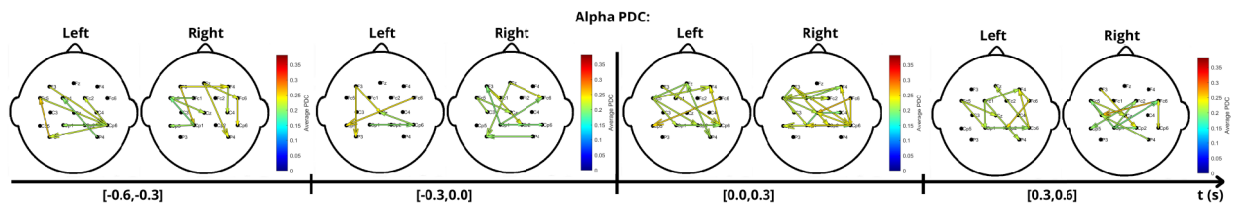
Fig. 1. Grand average of ERDS maps computed using the surface Laplacian during right-hand motor task execution. Each red line indicates the movement onset at time $t = 0$ for the corresponding EEG channel. The white line marks $t = -0.5$ s, providing a reference point to facilitate the interpretation of synchronization and desynchronization dynamics over time.

($p < 0.05$, corrected) when compared to the baseline fixation phase, and whose PDC values exceeded 50% of the maximum significant PDC within each condition, were displayed. Self-connections (predictive links from a channel to itself) were excluded, as they reflect signal autoregression rather than inter-regional interactions. These spatial patterns are shown in Figs. 2 and 3, corresponding to lateral and vertical grasping tasks, respectively. Notably, these maps reveal that the strongest PDC values tend to emerge during vertical grasping movements, though the timing of these peaks varies across frequency bands. For completeness, all frequency bands and conditions are fully depicted in the Supplementary Material.

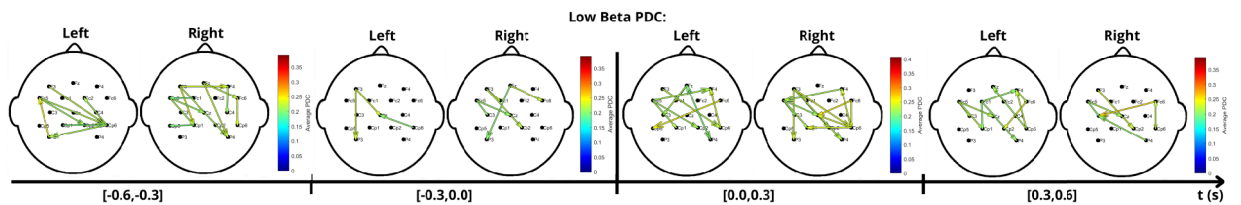
A quantitative summary of the EEG connectivity patterns is presented in Table 3. This Table reports the total number of statistically significant connections relative to the attention phase, for reach-and-grasp movements performed with either the left or right upper limb, as chosen by the subject. The data are shown across the analyzed frequency bands and at each of the four evaluated time intervals (two during the premovement phase and two during execution). In addition to the total number of significant connections, the Table reports how these are subdivided into interhemispheric connections, those originating in the left hemisphere and terminating in the right, and vice versa, as well as intrahemispheric connections, which both originate and terminate in the same hemisphere. All values are aggregated per experimental condition, as defined in the intracortical connectivity analysis described in the Methods section.

In general, a temporal modulation in the number of significant connections was observed as the onset of movement approached, with notable changes occurring during the $[0.3, 0]$ s and $[0, 0.3]$ s windows. This modulation was predominantly reflected as an increase across all frequency bands, especially during right-hand movements. Conversely, a reduction in connectivity was identified during vertical grasps performed with the left hand, suggesting a distinct pattern of cortical interaction specific to this condition.

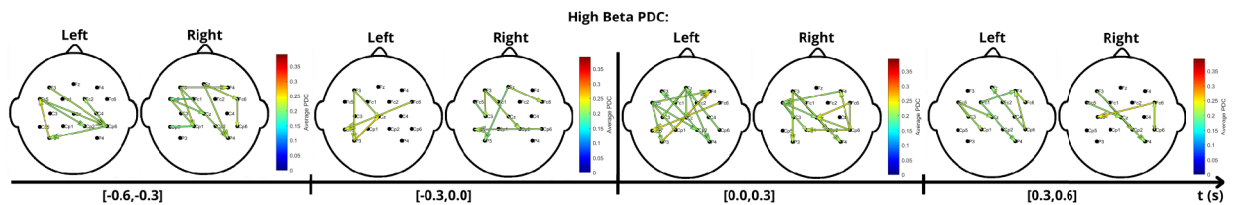
A higher number of intracortical L→L connections (regardless of frequency band) was found in comparison to R→R connections. This pattern was present during the initial three time windows $[0.6, 0.3]$, $[0.3, 0]$, and $[0,$



(a) Alpha band (8–12 Hz): Statistically significant connections observed during lateral grasp movements across temporal windows, relative to baseline during the attentional phase.



(b) Low Beta band (12–20 Hz): Statistically significant connections observed during lateral grasp movements across temporal windows, relative to baseline during the attentional phase.



(c) High Beta band (20–30 Hz): Statistically significant connections observed during lateral grasp movements across temporal windows, relative to baseline during the attentional phase.

Fig. 2. Statistically significant surface Laplacian EEG connectivity during lateral grasp movements across frequency bands. This Figure shows effective connections that exhibited a statistically significant difference in Partial Directed Coherence (PDC) ($p < 0.05$) during lateral grasp movements performed with both the right and left upper limb, relative to a baseline window during the attentional phase. The analysis was conducted on EEG signals transformed using the surface Laplacian. For the three frequency bands shown – Alpha (8–12 Hz), Low Beta (12–20 Hz), and High Beta (20–30 Hz) – only connections exceeding 50% of the maximum PDC value for each condition are displayed. Subfigures A–C illustrate the connectivity patterns at distinct stages of movement execution.

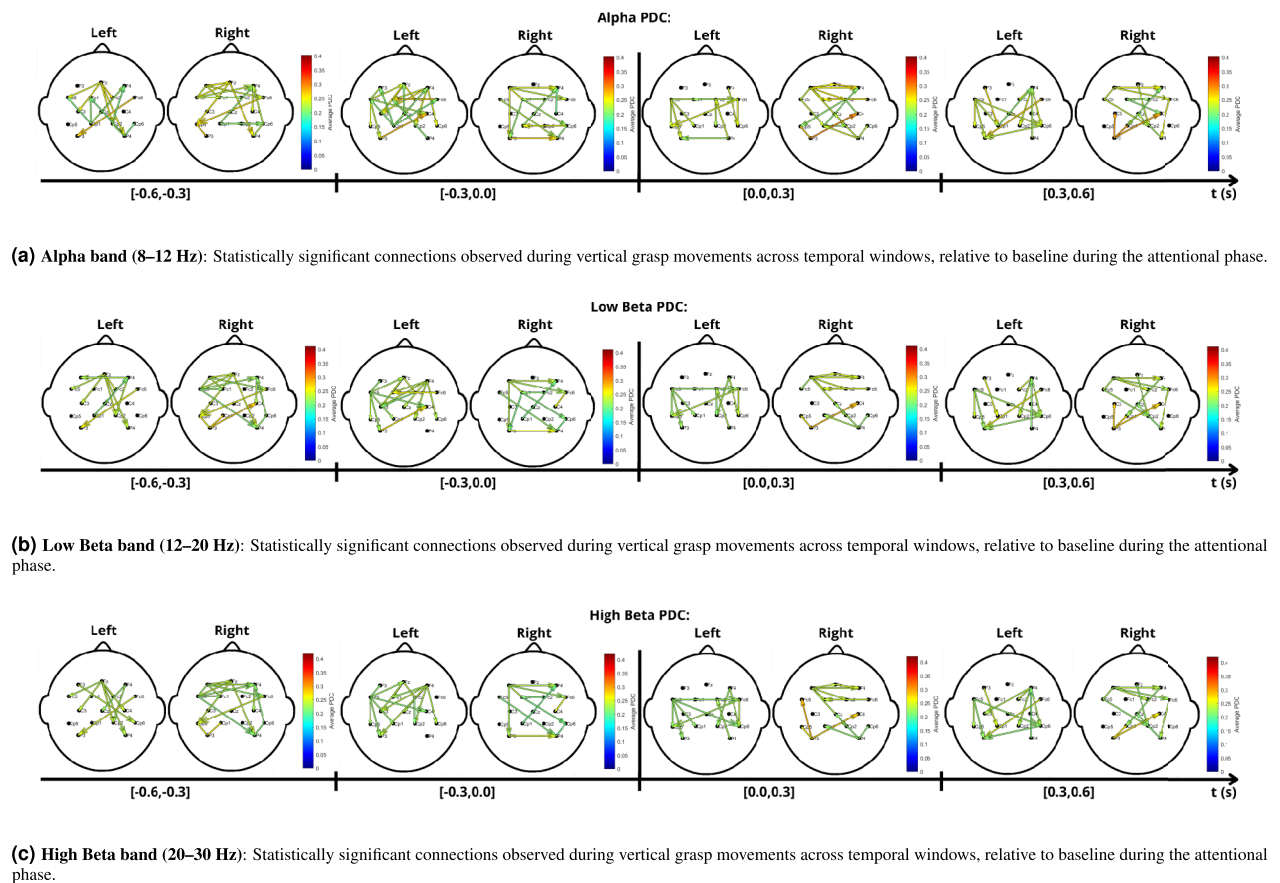


Fig. 3. Statistically significant surface Laplacian EEG connectivity during vertical grasp movements across frequency bands. This Figure shows functional connections that exhibited a statistically significant increase in Partial Directed Coherence (PDC) ($p < 0.05$) during vertical grasp movements performed with both the right and left upper limb, relative to a baseline window during the attentional phase. The analysis was conducted on EEG signals transformed using the surface Laplacian. For the three frequency bands shown – Alpha (8–12 Hz), Low Beta (12–20 Hz), and High Beta (20–30 Hz) – only connections exceeding 50% of the maximum PDC value for each condition are displayed. Subfigures A–C illustrate the connectivity patterns at distinct stages of movement execution.

0.3] s for lateral grasp movements. For vertical grasps, however, the predominance of L→L connections was restricted to the [0.3, 0] and [0, 0.3] s intervals. Concerning interhemispheric connectivity, L→R connections were more frequently observed than R→L, particularly within the premovement window [0.3, 0] s, where this asymmetry was consistent across both grasp types. Although variations in intracortical connection prevalence were also detected during the [0.6, 0.3] and [0, 0.3] s windows, the [0.3, 0.6] s interval was primarily characterized by a dominance of L→R connections.

To evaluate whether certain connectivity pathways reflect task-related processes independent of the limb used, a comparative analysis was conducted to identify directed connections consistently observed during movements performed with both the left and right upper limbs, within each grasping condition. The results are summarized in Table 4, which presents the connections shared across both lateral and vertical grasping movements for each time window and frequency band. The Table reveals that the highest number of consistent interlimb connections was observed during the early movement execution phase ([0, 0.3] s), across all frequency bands and both types of reach-and-grasp movements. Furthermore, a differential pattern was observed between the grasp types: in the three time windows from movement preparation to post-initiation (i.e., [−0.3, 0.6] s), lateral movements exhibited a higher number of shared connections between the limbs compared to vertical movements, regardless of the frequency band.

In particular, Cp1 was found to play a prominent role during the premovement phase ([−0.3, 0] s), establishing interhemispheric connections with Cp2 during lateral grasping and with F4 during vertical grasping. Moreover, for vertical grasping, a greater number of interlimb shared connections were observed during the early preparation phase ([−0.6, −0.3] s) compared to the immediate premovement window ([−0.3, 0] s). During early movement execution ([0, 0.3] s), several connections were consistently maintained across both limbs, and these were specific to the type of grasping movement performed. For lateral grasping, the shared connections included C3 → F4, C4 → F4, Fc5 → Cp1, Cz → Cp1, and Cp1 → Cp6. In contrast, for vertical grasping, the shared connections comprised Cp2 → F3, Fc2 → Fc1, Cz → Cp6, and Cp6 → P3. These consistent patterns were

Movement	Frequency band	Limb	[-0.6, -0.3] s				[-0.3, 0] s				[0, 0.3] s				[0.3, 0.6] s			
			Total	L→R	R→R	L→L	R→L	L→R	R→L	L→L	Total	L→R	R→L	L→L	Total	L→R	R→L	L→L
			Left	13	3	5	2	3	13	5	2	2	1	23	5	3	3	5
Lateral Grasp	Theta [4,8] Hz	Right	21	3	3	7	3	24	8	5	3	3	3	34	8	6	5	4
	Alpha (8,12] Hz	Left	11	3	4	1	3	12	4	2	2	1	1	25	5	3	4	2
		Right	20	2	3	7	3	20	6	5	3	2	2	35	10	6	5	4
	Beta Low (12,20] Hz	Left	12	4	4	1	3	12	4	2	2	1	1	27	5	3	5	3
		Right	21	3	3	7	3	22	6	5	4	2	2	34	10	5	4	5
	Beta High (20,30] Hz	Left	12	4	5	1	2	13	4	3	1	1	1	31	7	4	4	3
		Right	20	3	3	7	2	21	6	4	5	2	2	35	9	5	6	4
	Gamma > 30 Hz	Left	14	5	4	2	3	16	6	3	2	1	1	29	6	4	3	4
		Right	23	5	3	7	3	20	6	4	4	3	3	34	8	5	5	4
	Theta [4,8] Hz	Left	20	3	4	2	5	33	6	6	7	6	6	20	3	5	4	5
Vertical Grasp		Right	28	8	3	6	7	15	7	3	3	2	2	32	9	8	5	4
	Alpha (8,12] Hz	Left	19	2	4	2	5	34	6	6	8	6	6	18	1	5	4	5
		Right	29	8	3	6	8	16	7	3	3	3	3	32	9	7	3	5
	Beta Low (12,20] Hz	Left	22	3	6	2	4	36	6	6	9	7	7	20	1	5	4	5
		Right	34	10	4	7	8	15	7	2	3	3	3	30	8	7	3	6
	Beta High (20,30] Hz	Left	22	4	4	2	4	34	5	5	8	8	8	21	2	5	4	7
		Right	34	10	4	7	8	16	7	2	3	4	4	31	8	8	6	3
	Gamma > 30 Hz	Left	23	4	5	3	4	33	4	5	9	7	7	22	2	5	6	7
		Right	32	10	3	8	6	17	8	2	3	4	4	35	8	9	7	5

Table 3. Summary of significant PDC-based intracortical connections by frequency band, limb used, and time window relative to movement onset. Columns indicate the total number of connections (Total), as well as the directionality and hemispheric distribution: interhemispheric connections from left to right hemisphere (Inter: L→R) and from right to left (Inter: R→L), and intrahemispheric connections within the left hemisphere (Intra: L→L) and within the right hemisphere (Intra: R→R).

Movement	Frequency band	[−0.6, −0.3] s	[−0.3, 0] s	[0, 0.3] s	[0.3, 0.6] s
Lateral	Theta [4,8] Hz			C3 → F4	
		–	C3 → Fc6, Cp1 → Cp2	C4 → F4, Cp6 → F4, C4 → Fc1	C3 → F4, C4 → F4, C3 → Fc6
			Cp1 → Cp6, C3 → P4	Cz → C4, Fc5 → Cp1, Cz → Cp1	Fc5 → Cz, Cp1 → Cp2
				C4 → Cp6, Cp1 → Cp6	
	Alpha (8,12] Hz			C3 → F4, C4 → F4	
		–	C3 → Fc6, Cp1 → Cp2	Cp6 → F4, C4 → Fc1, Cz → C4	C3 → F4, C4 → F4, C3 → Fc6
				Fc5 → Cp1, Cz → Cp1, F3 → Cp6	Fc5 → Cz, Cp1 → Cp2
				C4 → Cp6, Cp1 → Cp6	
	Beta Low (12,20] Hz			C3 → F4, C4 → F4	C3 → F4
		F3 → F3	C3 → Fc6, Cp1 → Cp2	Cp6 → F4, C4 → Fc1, C3 → C3	C4 → F4, C3 → Fc2, C3 → Fc6
				Cz → C4, Fc5 → Cp1, Cz → Cp1	Fc5 → C3, Fc5 → Cz
				F3 → Cp6, C4 → Cp6, Cp1 → Cp6	Cp1 → Cp2
	Beta High (20,30] Hz			C3 → F4, C4 → F4	C3 → F4
		F3 → F3	F3 → F3, C3 → Fc6	Fc5 → Fc5, C4 → Fc1, C3 → C3	C4 → F4, Fc5 → Fc5, C3 → Fc2
			Cp1 → Cp2	Cz → C4, Fc5 → Cp1, Cz → Cp1	C3 → Fc6, Fc5 → C3, C3 → C3
				Cp5 → Cp6, Cp1 → Cp6	Fc5 → Cz
	Gamma >30 Hz			C3 → F4	
		P3 → Cp6	C3 → Fc6, Cp1 → Cp2	C4 → F4, C4 → Fc1, Cz → Cp5	C4 → F4, Fc5 → Fc5, Fc1 → Fc6
				Fc5 → Cp1, Cz → Cp1, Cp5 → Cp6	C3 → Fc6, Fc5 → C3, Fc5 → Cz
				Cp1 → Cp6	
Verical	Theta [4,8] Hz	Cp1 → F4, Cp2 → F4	Cp1 → F4	Cp2 → F3, Fc2 → Fc1, F3 → Fc2	F3 → F3, Cp1 → F4
		Fz → Fc5, Fz → P4		Fc5 → Cp1, Cz → Cp6, Cp2 → P3	
	Alpha (8,12] Hz	Cp1 → F4, Cp2 → F4	Cp1 → F4	Cp2 → F3, Fc2 → Fc1, Fc5 → Cp1	F3 → F3, Cp1 → F4
		Fz → Fc5, Fz → P4		Cz → Cp6, Cp2 → P3, P3 → P3	
	Beta Low (12,20] Hz	Cp2 → F3, Cp1 → F4, Cp2 → F4	Cp1 → F4	Cp2 → F3, Fc2 → Fc1	F3 → F3, Cp1 → F4
		Fz → Fc5, Fz → P4		Cz → Cp6, Cp2 → P3	Cp1 → P3
	Beta High (20,30] Hz	Cp1 → F4, Fz → Fc5	Cp1 → F4	Cp2 → F3, Fc2 → Fc1	F3 → F3, Cp1 → F4, Cz → Cz
		Fz → P4		Cz → Cp6, Cp2 → P3	Cz → Cp6, Cp1 → P3
	Gamma >30 Hz	Cp1 → F4, Fz → Fc5	–	Cp2 → F3	
				C4 → F4, P3 → Fc5, Fc2 → Fc1	Cp1 → F4, Cz → Cz
				Cz → Cp6, Cp2 → P3	Cz → Cp6, Cp1 → P3
				P4 → P3	

Table 4. Common connections observed between left and right upper limb movements for each grasp type, frequency band, and time window. Only connections that were statistically significant and common to both upper limbs are reported. The channels shown correspond to the surface Laplacian transform of the original signals.

observed across all frequency bands. Notably, in the context of early movement execution, the central region electrode Cz exhibited statistically significant outgoing connections toward Cp1 and Cp6, with the specific target depending on the type of movement executed by the participant.

To evaluate the temporal evolution of intracortical connectivity, an analysis was conducted to identify connections that remained stable across consecutive time windows. Laplacian-transformed EEG signals were used instead of raw EEG. Table 5 summarizes the connections that were preserved across the three temporal transitions: premovement, movement onset, and movement execution.

It was observed that, across the four analyzed time windows, certain connections remained consistent for each frequency band and appeared to be specific to both the type of movement and the limb involved in its execution.

For the lateral movement performed with the left upper limb, the connections C4 → F3 and Cp1 → Cp6 were statistically significant across all frequency bands and all time windows. Additionally, the connection C4 → P4 was consistently present across all time windows in the alpha and low beta bands, while in the gamma band, the connection Cp2 → Cp6 was preserved throughout all windows.

For the lateral movement performed with the right upper limb, the connection C4 → Fc1 was consistently observed across all frequency bands and all time windows. In the theta band, the connection C3 → Cp2 was also maintained, while in the theta, alpha, and low beta bands, the connection Fc5 → Cz appeared consistently but was absent in the high beta and gamma bands. In these latter bands, a different connection, C3 → Fc6, emerged. The connection Cp2 → Cp6 was observed exclusively in the gamma band.

For the vertical movement, no connection was found to be consistently present across all frequency bands and all four time windows. In the case of the vertical movement performed with the left upper limb, no connection remained stable across the four time points in the theta and alpha bands. From the low beta band onward,

Movement	Freq. band	Limb	[-0.6,-0.3] s → [-0.3,0] s	[-0.3,0] s → [0,0.3] s	[0,0.3] s → [0.3,0.6] s
Lateral	Theta [4,8] Hz	Left	C4 → F3, Cp1 → Cp6	C4 → F3, C3 → F4	C4 → F3, C3 → F4, C4 → F4, Cp6 → F4, C3 → C3
			C4 → P4	Fc6 → Cp5, Cp1 → Cp6	Cz → C4, Fc5 → Cp1, C3 → Cp1, Cp1 → Cp6
		Right	C4 → Fc1, Fz → Fc6, Cp2 → C3, Fc5 → Cz	Fc1 → F4, C4 → F4, F3 → Fc1, C4 → Fc1, C3 → Fc6, Fc5 → C3, C4 → C3	C3 → F4, C4 → F4, C4 → Fc1, C3 → Fc6, Fc5 → C3, Fc5 → Cz
			Cp1 → Cp1, C3 → Cp2, Cp2 → Cp6, C3 → P4	Fc5 → Cz, C3 → Cz, Cz → Cz, F3 → Cp2, C3 → Cp2, Cp1 → Cp6, C3 → P4	C3 → Cz, F3 → Cp2, C3 → Cp2, Fc6 → Cp6, P4 → P4
	Alpha (8,12] Hz	Left	C4 → F3, Cp1 → Cp6	C4 → F3, C3 → F4, C3 → C3	C4 → F3, C3 → F4, C4 → F4, Cp6 → F4, C3 → C3
			C4 → P4	Fc6 → Cp5, Cp1 → Cp6, C4 → P4	Cz → C4, Fc5 → Cp1, C3 → Cp1, Cp1 → Cp6, C4 → P4
		Right	C4 → Fc1, Fz → Fc6, Cp2 → C3, Fc5 → Cz	C4 → F4, F3 → Fc1, C4 → Fc1, C3 → Fc6, Fc5 → C3, C4 → C3	C3 → F4, C4 → F4, C4 → Fc1, C3 → Fc6, Fc5 → C3
			Cp1 → Cp1, Cp2 → Cp6, C3 → P4	Fc5 → Cz, C3 → Cz, Cz → Cz, F3 → Cp2, C3 → Cp2, C3 → P4	Fc5 → Cz, C3 → Cz, F3 → Cp2, C3 → Cp2, Fc6 → Cp6, P4 → P4
	Beta Low (12,20] Hz	Left	F3 → F3, C4 → F3	C4 → F3, C3 → F4, C3 → C3	C4 → F3, C3 → F4, C4 → F4, Cp6 → F4, Cp2 → Fc2
			Cp1 → Cp6, C4 → P4	C4 → C4, Cp1 → Cp6, C4 → P4	C3 → C3, Cz → C4, Fc5 → Cp1, C3 → Cp1, Cp1 → Cp6, C4 → P4
		Right	C3 → Fc5, C4 → Fc1, Fz → Fc6	C4 → F4, F3 → Fc1, C4 → Fc1, C3 → Fc6, Fc5 → C3, C4 → C3	C3 → F4, C4 → F4, C4 → Fc1, Fc6 → Fc6, C3 → Fc6, Fc5 → C3
			Cp2 → C3, Fc5 → Cz, Cp2 → Cp6, C3 → P4	Fc5 → Cz, C3 → Cz, Cz → Cz, F3 → Cp2, C3 → Cp2, C3 → P4	Fc5 → Cz, C3 → Cz, F3 → Cp2, C3 → Cp2, Fc6 → Cp6, Cp5 → Cp6
	Beta High (20,30] Hz	Left	F3 → F3, C4 → F3	C4 → F3, C3 → F4, C3 → C3	C4 → F3, Cp6 → Fz, C3 → F4, C4 → F4, Fc5 → Fc5
			C4 → C4, Cp1 → Cp6	C4 → C4, Fc6 → Cp5, Cp1 → Cp6, P3 → P3	Cp2 → Fc2, C3 → C3, Cz → C4, Fc5 → Cp1, C3 → Cp1, Cp1 → Cp6
		Right	F3 → F3, C3 → Fc5, C4 → Fc1	C4 → F4, F3 → Fc1, C4 → Fc1, C3 → Fc6	C3 → F3, Cz → F3, C3 → F4, C4 → F4, Fc5 → Fc5, C4 → Fc1, C3 → Fc2, Fc6 → Fc6
			Fz → Fc6, C3 → Fc6, Cp2 → C3, Cp2 → Cp6	Fc5 → C3, C3 → Cz, F3 → Cp2, C3 → Cp2, Cp2 → Cp6	C3 → Fc6, Fc5 → C3, C3 → C3, Fc5 → Cz, Cz → Cz, F3 → Cp2, C3 → Cp2, Fc6 → Cp6, Cp5 → Cp6, F3 → P4
	Gamma >30 Hz	Left	C4 → F3, C4 → C4	C4 → F3, C3 → F4	C4 → F3, Cp6 → Fz, C3 → F4, C4 → F4
			Cp1 → Cp6	C4 → C4, Fc6 → Cp5, Cp1 → Cp6	Cp2 → Fc2, Fc5 → Cp1, C3 → Cp1, Cp1 → Cp6
		Right	F3 → F3, F3 → F4, C3 → Fc5	C4 → F4, F3 → Fc1, C4 → Fc1, C3 → Fc6	Cz → F3, C4 → F4, Fc5 → Fc5, C4 → Fc1, C3 → Fc2, C3 → Fc6, Fc5 → C3
			C4 → Fc1, C3 → Fc6, Cp2 → C3, Cp2 → Cp6	C3 → Cz, Cz → Cp5, Cp6 → Cp5, Cp2 → Cp6	C3 → C3, Fc5 → Cz, C3 → Cp2, Fc6 → Cp6, Cp5 → Cp6, F3 → P4
Vertical	Theta [4,8] Hz	Left	Cp1 → F4, Cp2 → F4, Cp1 → Fc5, P4 → Fc1	Cp5 → F3, Fc5 → Cp5, Fc5 → Cp1	Cp5 → F3, Cp2 → F3, Fc5 → Cp5
			Fz → Fc2, Fz → C3, Cp5 → C4, F4 → Cp2, C4 → P4	F4 → Cp6, Fc5 → P3, C4 → P4	Fc5 → Cp1, F4 → Cp6, Cz → Cp6
		Right	C4 → F4, Cp1 → F4, F3 → Fc1	F3 → F4, C4 → F4, Cp1 → F4	F3 → F4, C4 → F4, Cp1 → F4, C4 → Fc1, C4 → Fc2
			F3 → Fc6, C4 → Cp1, C4 → Cp2, Cp5 → Cp6	F3 → Fc1, Fc2 → Fc1, C4 → Cp1	P4 → Cz, Cz → C4, P3 → C4, C4 → Cp1, C3 → P4, Cz → P4
	Alpha (8,12] Hz	Left	Cp1 → F4, Cp2 → F4, Cp1 → Fc5, P4 → Fc1	Cp5 → F3, Fc5 → Cp5, Fc5 → Cp1	Cp5 → F3, Cp2 → F3, Fc5 → Cp5, Fc5 → Cp1
			Fz → Fc2, Fz → C3, Cp5 → C4, F4 → Cp2, C4 → P4	F4 → Cp6, Fc5 → P3, C4 → P4	F4 → Cp6, Cz → Cp6, Fc5 → P3, Cp2 → P3
		Right	C4 → F4, Cp1 → F4, F3 → Fc1	F3 → F4, Cp1 → F4	F3 → F4, Cp1 → F4, F3 → Fc5, C4 → Fc1, C4 → Fc2
			F3 → Fc6, C4 → Cp1, C4 → Cp2, Cp5 → Cp6	F3 → Fc1, Fc2 → Fc1, P4 → Cp6	P4 → Cz, Cz → C4, P3 → C4, C3 → P4, Cz → P4
	Beta Low (12,20] Hz	Left	F4 → F4, Cp1 → F4, Cp2 → F4, Cp1 → Fc5, Cp1 → Fc1, P4 → Fc1	Cp5 → F3, Fc5 → Cp5, Fc5 → Cp1	Cp5 → F3, Cp2 → F3, Cz → Fc2, Cz → Cz, Fc5 → Cp5
			Fz → Fc2, Fz → C3, F4 → Cz, Cp5 → C4, F4 → Cp2, C4 → P4	F4 → Cp2, F4 → Cp6, Fc5 → P3, C4 → P4	F4 → Cp2, F4 → Cp6, Cz → Cp6, Fc5 → P3, Cp2 → P3
		Right	Cp1 → F4, F3 → Fc1, F3 → Fc6	F3 → F4, Cp1 → F4	F3 → F3, F3 → F4, Cp1 → F4, F3 → Fc5, C4 → Fc1, F3 → Fc2
			C4 → Cp1, C4 → Cp2, Cp5 → Cp6, Fc5 → P4	F3 → Fc1, Fc2 → Fc1, P4 → Cp6	C4 → Fc2, P4 → Cz, Cz → C4, P3 → C4, Cp5 → P3, Cz → P4
	Beta High (20,30] Hz	Left	F4 → F4, Cp1 → F4, Cp2 → F4, Cp1 → Fc5, Cp1 → Fc1, P4 → Fc1	Cp5 → F3, Fc5 → Cp5, Fc5 → Cp1, F4 → Cp2	Cp5 → F3, Cp2 → F3, Cz → Fc2, Cz → Cz, Fc5 → Cp5
			Fz → Fc2, Fz → C3, F4 → Cz, Cp5 → C4, F4 → Cp2, C4 → P4	Fc2 → Cp2, F4 → Cp6, Fc5 → P3, C4 → P4	F4 → Cp2, F4 → Cp6, Fc5 → Cp6, Cz → Cp6, Fc5 → P3, Cp2 → P3
		Right	F3 → F4, Cp1 → F4, F3 → Fc1, C4 → Fc2	F3 → F4, Cp1 → F4, F3 → Fc1	F3 → F3, F3 → F4, Cp1 → F4, C4 → Fc1, F3 → Fc2, C4 → Fc2, P4 → Cz
			F3 → Fc6, C4 → Cp1, C4 → Cp2, Cp5 → Cp6, Fc5 → P4	Fc2 → Fc1, C4 → Fc2, P4 → Cp6	C4 → Fc2, Cz → C4, Cz → Cp6, Cp6 → Cp6, P4 → Cp6, Cp5 → P3, Cz → P4

Table 5. Intracortical connections preserved across consecutive time windows are shown for each frequency band and motor condition. For each grasp type (lateral or vertical) and limb (left or right), connections that remained significant between adjacent time intervals were identified. The direction of sustained effective connectivity between cortical regions was estimated using Partial Directed Coherence and is indicated by arrows.

		Outdegree				Indegree			
		$t = [-0.6, -0.3]$	$t = [-0.3, 0]$	$t = [0, 0.3]$	$t = [0.3, 0.6]$	$t = [-0.6, -0.3]$	$t = [-0.3, 0]$	$t = [0, 0.3]$	$t = [0.3, 0.6]$
Left	Theta	C4,Cp6	C3,F3	C4,Cz	C3,Fc1	Cp6,F3	Cz,F3	Fz,F4	F4,Cp1
	Alpha	C4,F3	C3,F3	C4,Cz	C3,Fc1	Cp6,F3	Cz,F3	Fz,F4	F4,Cp1
	Low beta	C4,F3	C3,F3	C4,Fz	C3,Fc5	Cp6,F3	F3,Cz	Fz,F4	F4,C3
	High beta	C4,F3	F3,C3	C4,Fz	C3,Fc5	Cp6,F3	Cz,F3	Fz,F4	F4,C3
	Gamma	C4,Fc5	F3,C3	C4,Fz	C3,Fc5	F4,Cp6	Cz,P4	Fz,F4	F4,Fc6
Right	Theta	C3,Fz	Cp1,C3	C3,C4	C3,Cp1	Cp1,Fc5	F4,Fc6	Cz,F4	Fc5,Fc6
	Alpha	Fz,C3	C3,C4	C3,Cz	C3,Cp1	Cp1,Fc5	Fc6,C3	Cz,F4	Fc5,Fc6
	Low beta	F3,C4	C3,C4	C3,Cz	C3,Cp1	Cp1,Fc5	Fc6,C3	Cz,Cp6	Fc6,Fc5
	High beta	F3,Fz	C3,F3	C3,Cz	C3,C4	Fc5,Cp1	Fc6,C3	Cz,Cp6	Fc5,Fc6
	Gamma	F3,Fc5	F3,C3	C3,Cz	C3,C4	Fc5,Cp6	Cp5,F4	Cp6,F4	Fc5,Cp6

Table 6. Outdegree and Indegree of significant PDC connections for each of the five analysed frequency bands during lateral grasp movements executed with both upper limbs.

		Outdegree				Indegree			
		$t = [-0.6, -0.3]$	$t = [-0.3, 0]$	$t = [0, 0.3]$	$t = [0.3, 0.6]$	$t = [-0.6, -0.3]$	$t = [-0.3, 0]$	$t = [0, 0.3]$	$t = [0.3, 0.6]$
Left	Theta	Fz,P4	F4,Fc5	Fc5,Cz	Fc5,Cp2	F4,Cp2	Fc1,C4	P3,Cp6	P3,Cp6
	Alpha	Fz,P4	F4,Fc5	Fc5,Cz	Fc5,Cp2	F4,Cp2	Fc1,C4	P3,Cp6	P3,Cp6
	Low beta	Fz,F4	F4,Fc5	Fc5,Fc2	Cp2,F4	F4,C3	Fc1,C4	P3,Fc2	P3,F4
	High beta	Fz,P4	F4,Fc5	Fc5,Fc2	F4,Cp2	F4,C3	Fc1,Fz	Cp6,P3	P3,F4
	Gamma	Fz,F4	F4,Fc5	Fc5,Fc2	F4,Cp2	C3,Cp6	Fc1,C4	Cp5,Cp6	P3,F4
Right	Theta	C4,F3	F3,C4	F3,C3	C4,F3	F4,Cp5	F4,Fc1	F4,Fc1	Cp1,F4
	Alpha	C4,F3	F3,C4	F3,C3	C4,F3	F4,Cp5	F4,Cp6	Fc1,Cp6	Cp1,F4
	Low beta	C4,Cp1	F3,Fc1	F3,Cp1	C4,F3	F4,Cp5	Cp6,F4	Fc1,Cp6	Cp1,F4
	High beta	C4,F3	F3,C4	F3,Fc2	C4,F3	F4,Cp5	Cp6,F4	Cp6,Fc1	Cp6,Cp1
	Gamma	C4,Cp1	F3,Fc1	F3,P4	C4,F3	F4,F3	Cp6,F4	Cp6,F4	Cp1,Cp6

Table 7. Outdegree and Indegree of significant PDC connections for each of the five analysed frequency bands during vertical grasp movements executed with both upper limbs.

the connection $F4 \rightarrow Cp2$ was consistently observed across the four time windows. In the gamma band, the connections $F4 \rightarrow Cp6$ and $Fc5 \rightarrow P3$ were present throughout all temporal windows.

For the vertical movement executed with the right upper limb, the connection $Cp1 \rightarrow F4$ was observed across all four time windows in the theta, alpha, low beta, and high beta bands. In the theta band specifically, two additional connections were found to be stable across all time windows: $C4 \rightarrow F4$ and $C4 \rightarrow Cp1$. In the gamma band, two connections— $F3 \rightarrow F4$ and $C4 \rightarrow Fc2$ —were preserved throughout all four time points.

In addition to the connections that remained stable across all time windows for each specific motor task condition, a subset of connections was identified that consistently appeared during the temporal transition associated with movement initiation. Specifically, these connections were present in both the premovement window $[-0.3, 0]$ s and the movement onset window $[0, 0.3]$ s, across all analyzed frequency bands. For the lateral movement performed with the left upper limb, the connection $C3 \rightarrow F4$ was preserved during this transition. In the case of lateral movement performed with the right upper limb, the following connections were identified: $C4 \rightarrow F4$, $F3 \rightarrow Fc1$, $C3 \rightarrow Fc6$, and $C3 \rightarrow Cz$. For the vertical movement executed with the left upper limb, the connections $Cp5 \rightarrow F3$, $Fc5 \rightarrow Cp5$, $Fc5 \rightarrow Cp1$, $F4 \rightarrow Cp6$, and $Fc5 \rightarrow P3$ were consistently present across the premovement and movement onset windows. Lastly, for the vertical movement performed with the right upper limb, the following connections were observed during the same transition: $F3 \rightarrow F4$, $F4 \rightarrow Fc1$, and $Fc2 \rightarrow Fc1$.

To assess the relative contribution of different cortical sites as either sources or recipients of directed information flow, out-degree and in-degree analyses were carried out. The corresponding results are summarized in Tables 6 and 7, which display the findings for lateral and vertical grasping conditions, respectively. In each case, the two electrodes with the largest number of outgoing links (out-degree), indicating that their activity predicts that of other sites, and the two electrodes with the largest number of incoming links (in-degree), suggesting that others can predict their activity, are reported.

For lateral grasping with the left hand, channel F3 emerged as a consistent driver of connectivity during the premovement phase across all frequency bands, accompanied by alternating contributions from C3 and C4. In terms of in-degree, F3 was also highlighted as relevant, while Cz became a prominent target of incoming connections immediately before movement onset ($t = [-0.3, 0]$). In contrast, right-hand movements did not reveal a dominant node equivalent to F3. Instead, the premovement stage involved a distributed contribution

from C3, C4, Fz, F3, and Cp1 in out-degree, whereas during execution C3 and C4 were the primary drivers of connectivity. For in-degree, Fc5 and Fc6 showed increased involvement during preparation, which decreased around movement onset ($t = [0, 0.3]$), before regaining importance in the late execution window.

The outcomes of the out-degree and in-degree analyses for vertical grasping are summarized in Table 7, which reports, for each time window and frequency band, the two electrodes with the highest number of outgoing links (out-degree) and the two with the highest number of incoming links (in-degree). Electrodes with elevated out-degree values can be regarded as sources whose activity predicts that of other sites, whereas electrodes with elevated in-degree values are interpreted as targets whose activity is predicted by other nodes.

In left-hand trials, frontal electrodes (notably Fz and F4) together with central sites (such as Fc5) consistently acted as sources of directed connectivity across temporal windows and frequency ranges. In contrast, the in-degree analysis revealed that posterior regions, including P3 and Cp6, became prominent targets of incoming connections, particularly during the execution stages of the movement. For right-hand trials, electrodes C4 and F3 emerged as dominant in the out-degree analysis, functioning as main sources of directed interactions across several bands and periods. Regarding in-degree, increased receptivity was observed in Fc5, Fc6, and Cp1 at different stages, indicating dynamic reorganizations of cortical integration depending on both movement phase and frequency band.

Corticomuscular connectivity dynamics during vertical and lateral grasping

The dynamics of corticomuscular connectivity were analyzed during vertical and lateral grasping movements to investigate whether effective connectivity differed depending on the limb used to execute the movement. EEG signals were Laplacian transformed at electrodes C3, Cz, and C4 to enhance spatial specificity, and connectivity was assessed across five frequency bands: theta (4–8 Hz), alpha (8–12 Hz), low beta (13–20 Hz), high beta (21–30 Hz), and gamma (>30 Hz), using rectified EMG signals recorded from the lower trapezius, anterior deltoid, and pronator teres muscles.

To evaluate statistical differences between conditions, a point-by-point Wilcoxon signed-rank test was applied to compare the PDC curves corresponding to each limb. To correct for multiple comparisons across time points, the resulting p -values were adjusted using the Benjamini Hochberg false discovery rate (FDR) procedure, with a significance threshold of $\alpha = 0.05$. The resulting average PDC curves for each frequency band are shown in Figs. 4 and 5, where red or blue shaded areas around the curves indicate ± 2 standard deviations. Gray shaded regions indicate time points with statistically significant differences between right and left limb movements (uncorrected), and green shaded regions highlight those that remained significant after FDR correction.

In the plots, it can be seen that the corticomuscular connectivity from EEG to EMG channels decreases during the movement preparation phase and remains low throughout the execution. No substantial differences were found between right and left upper limb executions, except for a few time points toward the end of the lateral grasping movement. Specifically, significant differences were observed between C3 and the Lower Trapezius in the theta, alpha, low beta, and high beta bands, and between C4 and the Lower Trapezius in the alpha, low beta, high beta, and gamma bands.

In the analysis of vertical grasping movements, some differences emerged during the premovement phase in the high beta and gamma bands, particularly in the connectivity between C3 and the Anterior Deltoid, as well as between C4 and the Anterior Deltoid. Additionally, at the end of the movement, differences were found between Cz and the Pronator Teres in the gamma band.

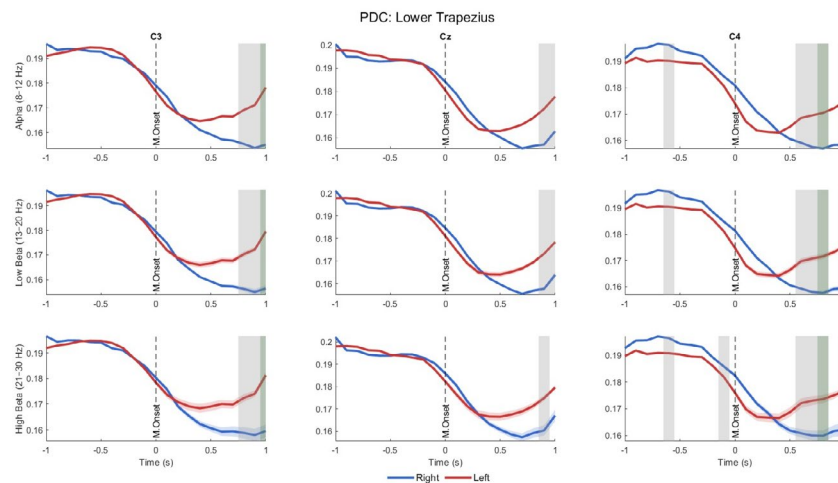
A complementary analysis was conducted to compare the EEG-to-EMG effective connectivity values obtained during the movement preparation and execution with those calculated during the fixation cross period. The Wilcoxon test revealed statistically significant differences emerging near movement onset, and the specific timing of these effects depended on the type of movement performed. Notably, significant differences were detected only in the left tail of the distribution, corresponding to lower PDC values during movement-related periods than during fixation.

Figures 6 and 7 illustrate the temporal distribution of significant differences for each muscle and for the Alpha (8–12 Hz), Low Beta (13–20 Hz), and High Beta (21–30 Hz) bands. Orange rectangles indicate significant differences for right-limb movements, blue rectangles for left-limb movements, violet for differences common to both limbs, and white when no significant difference was observed.

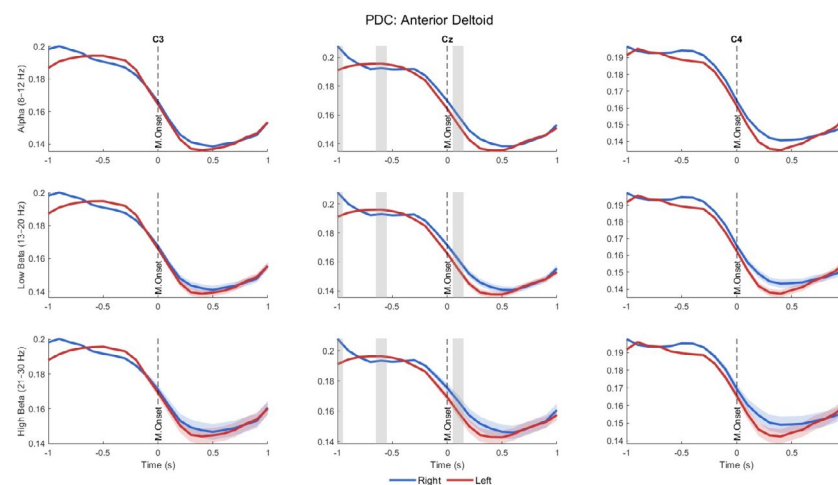
For lateral grasp movements, clear differences emerged between the two limbs in the Alpha (8–12 Hz), Low Beta (13–20 Hz), and High Beta (21–30 Hz) bands. For the right limb, significant reductions in EEG-to-EMG effective connectivity began at movement onset (0 s) and extended continuously up to +1 s for the Pronator Teres and Anterior Deltoid. These effects appeared earlier when analyzed from C3 and slightly later from Cz and C4, particularly in the High Beta band. The Lower Trapezius also showed continuous decreases starting near 0.1 s and persisting until movement completion.

For the left limb, reductions appeared later and were generally less continuous. For the Pronator Teres, significant differences were observed intermittently between 0.1 and 1 s in Alpha and Low Beta, with shorter, sporadic effects in High Beta. In contrast, the Anterior Deltoid showed a more sustained decrease, with continuous reductions starting around 0.1 s and extending until +1 s across Alpha, Low Beta, and High Beta bands. The Lower Trapezius exhibited a delayed and fragmented pattern, with reductions typically emerging between 0.2 and 0.4 s, especially in the High Beta band.

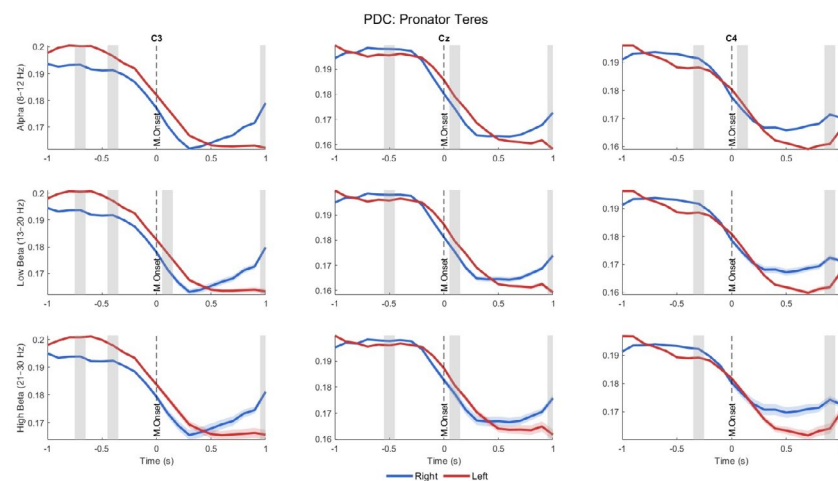
For vertical grasp movements, both limbs showed widespread and sustained reductions in EEG-to-EMG effective connectivity across the Alpha (8–12 Hz), Low Beta (13–20 Hz), and High Beta (21–30 Hz) bands. For the right limb, significant decreases began at movement onset (0 s) and persisted continuously up to +1 s for the Pronator Teres and Anterior Deltoid. These effects were most consistent in analyses from C3, while for Cz and C4 the reductions typically started around 0.1 s and remained stable throughout execution. The Lower Trapezius also showed a continuous decrease starting near 0 s in Alpha and Low Beta, and from 0.1 s in High Beta.



(a) Mean PDC from EEG to EMG signals of the Lower Trapezius (shaded areas indicate significant differences)



(b) Mean PDC from EEG to EMG signals of the Anterior Deltoids (shaded areas indicate significant differences)



(c) Mean PDC from EEG to EMG signals of the Pronator Teres (shaded areas indicate significant differences)

For the left limb, connectivity reductions followed a similar but slightly earlier pattern. For the Pronator Teres, significant and continuous decreases began at or just before movement onset (0.1 to 0 s) and extended up to +1 s across all three frequency bands and EEG channels. The Anterior Deltoid showed comparable continuous reductions starting around 0.1 s for C3 and C4, and near 0 s for Cz. The Lower Trapezius exhibited a stable decrease beginning at 0 s for C3 and Cz, and slightly later (around 0.1 s) for C4, consistent across the three frequency bands.

Taken together, these findings indicate that corticomuscular coupling in the Alpha and Beta bands consistently weakened around movement onset, reflecting a transient reduction in the prediction of muscle activity from

◀ **Fig. 4.** Corticomuscular effective connectivity during lateral grasp movements performed with the right or left upper limb. Partial Directed Coherence (PDC) was estimated from Laplacian-transformed EEG signals (C3, Cz, and C4) to rectified EMG signals from three upper limb muscles: (a) Lower Trapezius, (b) Anterior Deltoid, and (c) Pronator Teres. Connectivity was assessed for each limb across three frequency bands (Alpha, Low Beta, and High Beta) and during both the premovement and movement execution phases. Red or blue shaded areas around the curves indicate ± 2 standard deviations. Gray shaded regions mark time intervals with statistically significant differences between movements performed with the left versus the right upper limb, while green shaded regions denote differences that remained significant after false discovery rate (FDR) correction.

cortical signals during the transition from movement preparation to execution. Lateral grasps showed slightly earlier and more localized decreases, whereas vertical grasps involved broader and more symmetric reductions across EEG–EMG pairs. Supplementary figures illustrate the temporal evolution of significant differences across all five frequency bands (Theta, Alpha, Low Beta, High Beta, and Gamma) for each EEG channel and muscle.

Discussion

Voluntary reach-and-grasp movements rely on dynamic neural processes that evolve continuously from planning to execution^{1–12}. In this study, we assessed the temporal evolution of effective connectivity using Partial Directed Coherence (PDC), with a focus on both intracortical and corticomuscular interactions. While PDC has been widely applied in simulations^{29,31,48–51} and in clinical contexts^{34,52–55}, its use with real EEG and EMG signals during voluntary motor behavior remains limited, particularly in paradigms that capture the unfolding temporal dynamics of movement-related brain activity.

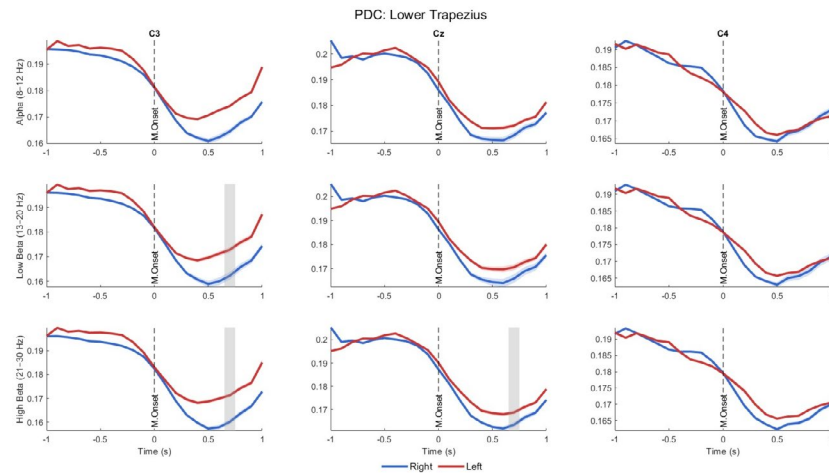
To address this gap, we implemented a movement task in which participants freely decided both when to initiate the movement and which limb to use. This setup engaged internal decision-making processes, enabling the analysis of connectivity patterns related not only to motor execution but also to action planning and intention. Beyond the classical findings of alpha/beta desynchronization and gamma synchronization, the key contribution of this study lies in applying a time-resolved PDC approach to self-initiated grasping movements. This framework revealed evolving directional interactions across pre-movement and movement phases, offering new insights into the temporal organization of effective connectivity. Our results suggest that decision-related activity may be reflected not only in event-related desynchronization/synchronization (ERD/ERS) patterns but also in the directional flow of information between cortical areas and between cortex and muscles, underscoring the dynamic and distributed nature of motor-related connectivity.

A surface Laplacian transformation was applied before both ERDS and connectivity analyses to improve the spatial specificity of EEG signals. This method enhances locally generated cortical activity by amplifying high spatial frequency components while attenuating widespread low-frequency signals linked to volume conduction and non-neuronal sources^{56–58}. It has been suggested that the Laplacian acts as an effective spatial filter and reduces electromyographic contamination by emphasizing local voltage gradients⁵⁸. While this preprocessing step is particularly useful for enhancing the reliability of ERDS estimation and supporting directional connectivity analyses, it primarily emphasizes short-range interactions and may reduce sensitivity to long-range interactions. Nevertheless, the observed interhemispheric and corticomuscular connectivity patterns in our study likely reflect genuine functional interactions rather than artifacts of volume conduction, as they remain robust across participants and conditions. The Laplacian was computed using the full set of 32 electrodes (excluding only channels rejected due to artifacts), and connectivity analyses were subsequently performed on a subset of 16 electrodes of interest, ensuring that the spatial filtering benefited from the higher-density electrode configuration.

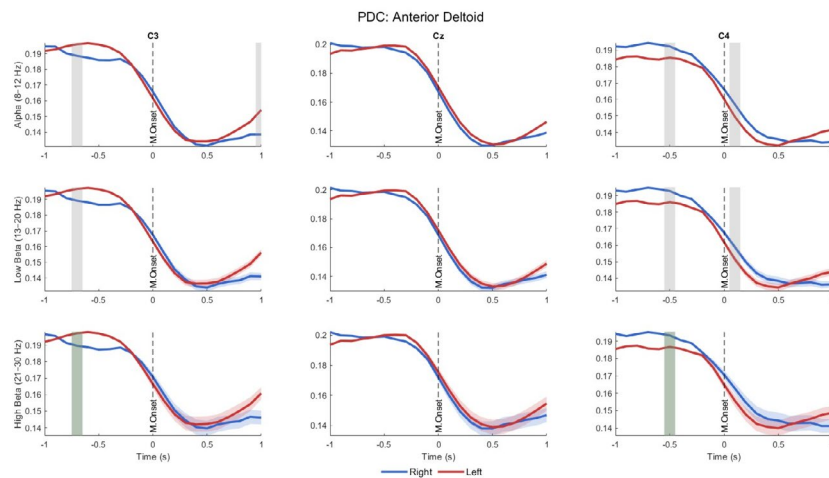
The analysis of event-related desynchronization/synchronization (ERDS) revealed consistent spectral power modulations during both the preparation and execution of self-initiated reach-and-grasp actions. A sustained desynchronization was observed in the alpha (8–12 Hz) and beta (12–30 Hz) bands, beginning approximately 0.5 seconds before movement onset and often persisting throughout the execution phase, consistent with previous studies^{59,60}. In the theta band (4–8 Hz), a transient synchronization emerged around 0.4 seconds before movement onset and reappeared toward the end of the analysis window, suggesting involvement not only in movement preparation but also in post-movement evaluation. This theta-band activity is hypothesized to reflect the engagement of neural processes underlying the internal decision to initiate movement and its subsequent monitoring, in line with previous studies linking theta synchronization to decision-making processes^{61–63}.

Additionally, gamma-band activity (>30 Hz) was observed across nearly all EEG channels, beginning approximately 0.3 seconds before movement onset, briefly diminishing after movement initiation, and in some cases re-emerging during late execution. The early onset of gamma activity is hypothesized to support the initiation and coordination of movement, potentially reflecting the binding of sensorimotor representations or the resolution of motor intentions, functions that have been associated with gamma oscillations in the context of motor planning and execution^{64–66}. Altogether, these ERDS patterns underscore the dynamic and frequency-specific modulation of cortical activity during voluntary motor behavior. The distinct temporal profiles of theta, alpha, beta, and gamma oscillations suggest that motor planning, decision-making, and execution are mediated by parallel but interacting neural processes distributed across the cortex.

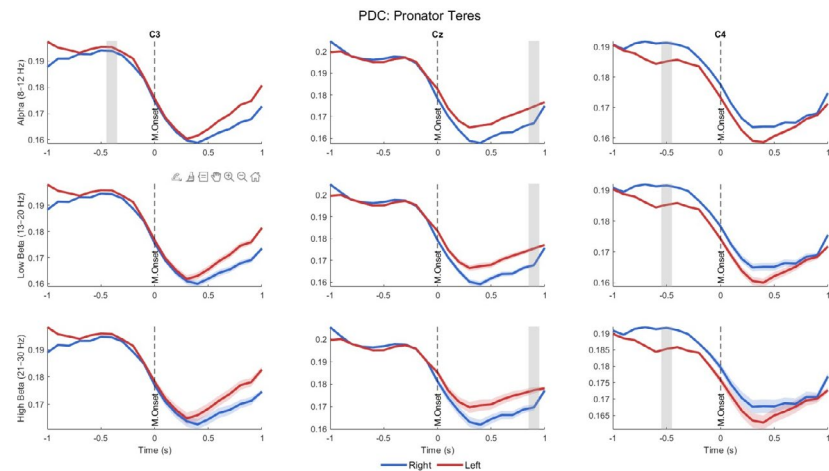
To further investigate the directional interactions underlying these processes, Partial Directed Coherence (PDC) was employed to estimate frequency specific information flow between signals. While PDC is a powerful tool for assessing effective connectivity in EEG and EMG data, its validity critically depends on methodological assumptions that are often overlooked in the literature. Chief among these are the requirement for local



(a) Mean PDC from EEG to EMG signals of the Lower Trapezius (shaded areas indicate significant differences)

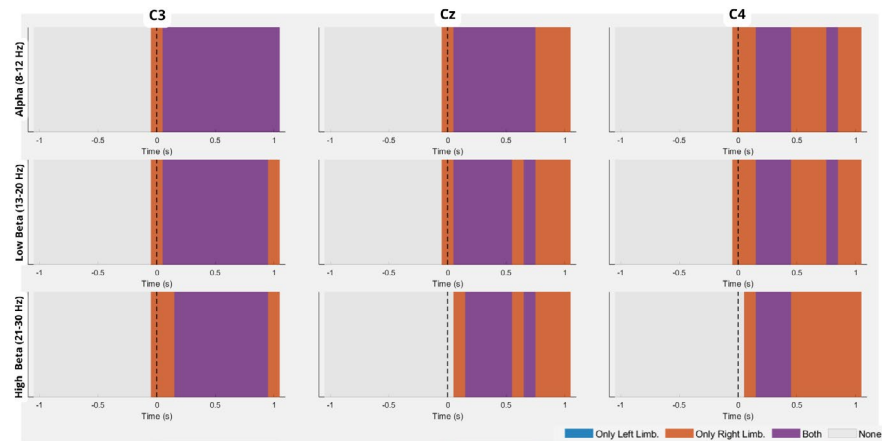


(b) Mean PDC from EEG to EMG signals of the Anterior Deltoid (shaded areas indicate significant differences)

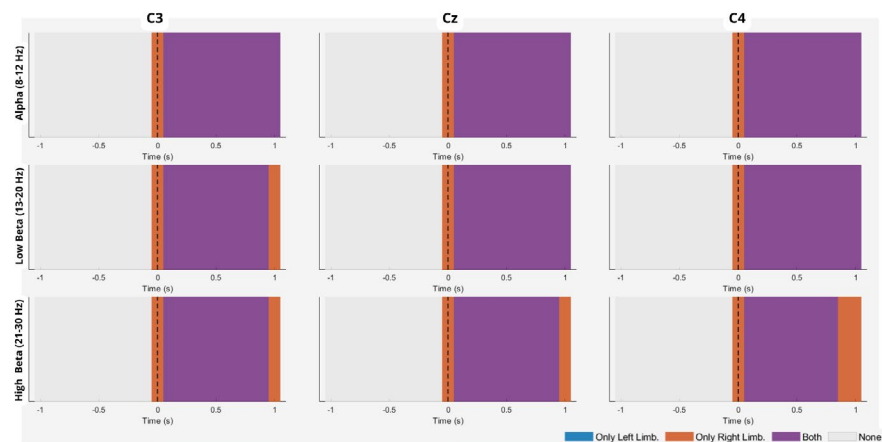


(c) Mean PDC from EEG to EMG signals of the Pronator Teres (shaded areas indicate significant differences)

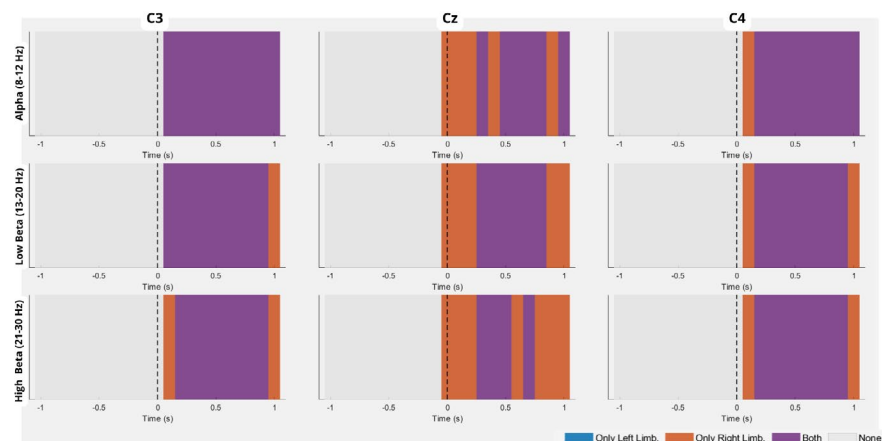
Fig. 5. Corticomuscular effective connectivity during vertical grasp movements performed with the right or left upper limb. Partial Directed Coherence (PDC) was estimated from Laplacian-transformed EEG signals (C3, Cz, and C4) to rectified EMG signals from three upper limb muscles: (a) Lower Trapezius, (b) Anterior Deltoid, and (c) Pronator Teres. Connectivity was assessed for each limb across three frequency bands (Alpha, Low Beta, and High Beta) and during both the pre-movement and movement execution phases. Red or blue shaded areas around the curves indicate ± 2 standard deviations. Gray shaded regions mark time intervals with statistically significant differences between movements performed with the left versus the right upper limb, while green shaded regions denote differences that remained significant after false discovery rate (FDR) correction.



(a) Pronator Teres: Time-resolved significant differences in corticomuscular connectivity relative to baseline across Alpha, Low Beta, and High Beta bands.

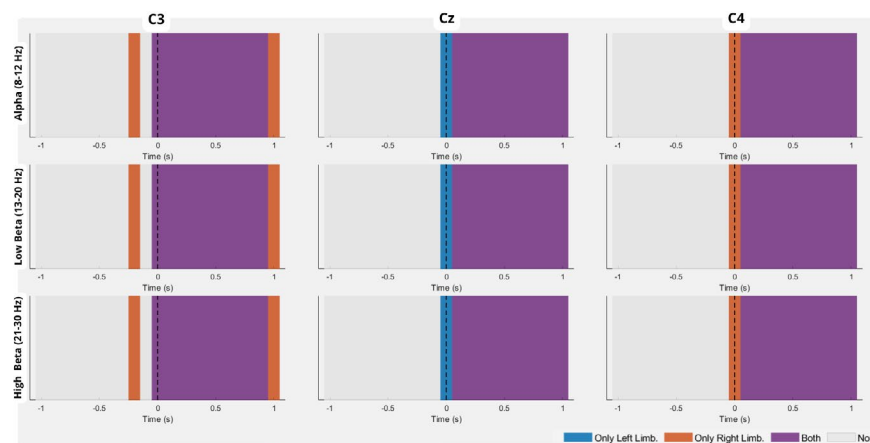


(b) Anterior Deltoid: Time-resolved significant differences in corticomuscular connectivity relative to baseline across Alpha, Low Beta, and High Beta bands.

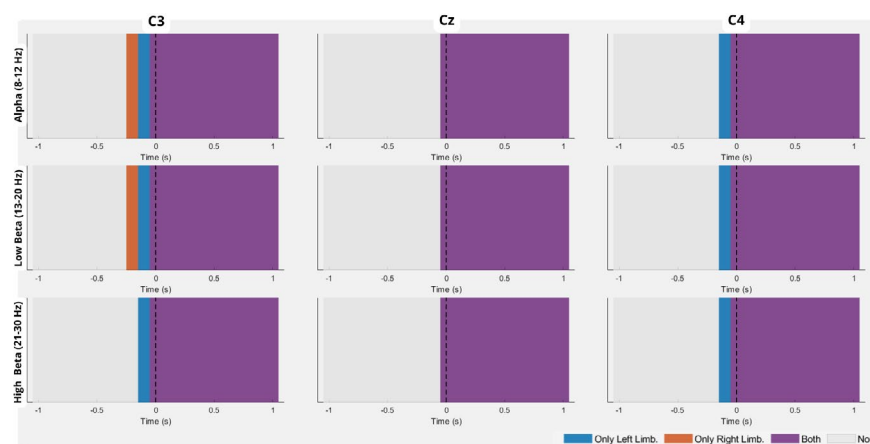


(c) Lower Trapezius: Time-resolved significant differences in corticomuscular connectivity relative to baseline across Alpha, Low Beta, and High Beta bands.

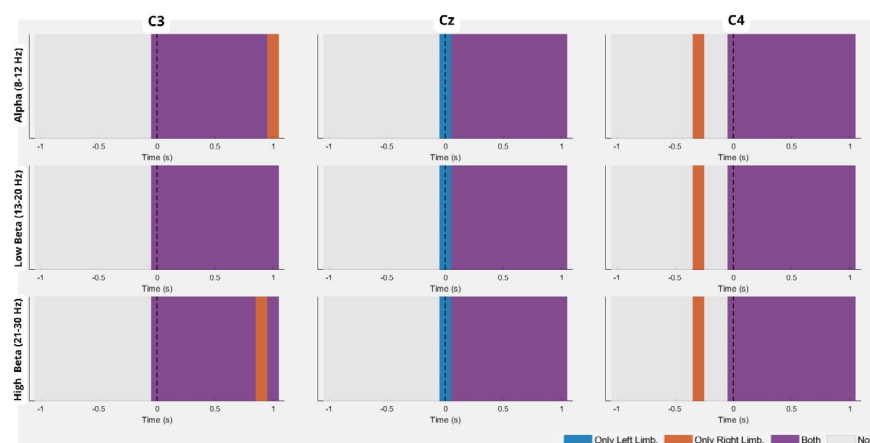
Fig. 6. Temporal distribution of significant differences relative to baseline in PDC assessed to evaluate corticomuscular connectivity during lateral grasp movements. Orange rectangles indicate significant differences for right limb movements, blue rectangles for left limb movements, violet for differences common to both limbs, and white when no difference is detected. Only Alpha (8–12 Hz), Low Beta (13–20 Hz), and High Beta (21–30 Hz) bands are shown.



(a) Lower Trapezius: Time-resolved significant differences in corticomuscular connectivity relative to baseline across Alpha, Low Beta, and High Beta bands.



(b) Anterior Deltoid: Time-resolved significant differences in corticomuscular connectivity relative to baseline across Alpha, Low Beta, and High Beta bands.



(c) Pronator Teres: Time-resolved significant differences in corticomuscular connectivity relative to baseline across Alpha, Low Beta, and High Beta bands.

Fig. 7. Temporal distribution of significant differences relative to baseline in PDC assessed to evaluate corticomuscular connectivity during vertical grasp movements. Orange rectangles indicate significant differences for right limb movements, blue rectangles for left limb movements, violet for differences common to both limbs, and white when no difference is detected. Only Alpha (8–12 Hz), Low Beta (13–20 Hz), and High Beta (21–30 Hz) bands are shown.

stationarity of the signals within the analysis window and the convergence of the multivariate autoregressive (MVAR) model. These assumptions are frequently taken for granted, even though violations can substantially distort connectivity estimates and lead to misleading interpretations.

In the present study, specific measures were taken to address these methodological challenges. The data were downsampled to 80 Hz for intracortical analyses and to 100 Hz for corticomuscular analyses, following the Nyquist criterion, thereby limiting the analyzable frequency range to 4–40 Hz and 4–50 Hz, respectively. This choice was aimed at improving model stability and convergence, while maintaining focus on the spectral components most relevant to motor processing. Importantly, analysis windows were chosen to maximize the likelihood of local stationarity, thereby supporting the validity of the MVAR modeling and the resulting PDC estimates. Model order was systematically determined using the Akaike Information Criterion (AIC) on a trial-by-trial basis, and stationarity was formally assessed with Augmented Dickey–Fuller (ADF) and KPSS tests. Although a small subset of channels did not fully meet the criteria, the short (300 ms) analysis windows and consistency with previous movement-related EEG studies support the assumption of local stationarity^{33–35,67}. Furthermore, MVAR convergence was ensured in at least 70% of trials per subject, with only one subject excluded in the corticomuscular analysis at -10 s. These methodological safeguards strengthen confidence in the robustness of the reported PDC estimates. These considerations underscore that, while evaluating signal properties and model behavior is essential for valid PDC estimates, such methodological transparency is rarely reported in previous studies, highlighting a gap in the literature.

In several previous studies on functional connectivity, movement execution has been evaluated, and increases in connectivity within the alpha and beta bands have been reported relative to the baseline state^{23–26}. These findings are consistent with the results presented in this study. Kim et al.²⁷ investigated connectivity during the Bereitschaftspotential. They found that during the desynchronization period, also identified in the ERDS analysis of the present study, the connections in the beta and gamma bands differed from baseline connectivity. When compared with our results, a similar pattern emerges; however, the present analysis also considered the theta and alpha bands, where changes in connectivity were likewise observed relative to the attention state in which participants were not engaged in any movement.

When the number of statistically significant connections was analyzed, it was observed that movements performed with the right upper limb were generally associated with a greater number of connections than those executed with the left upper limb, irrespective of whether the grasp was lateral or vertical. An exception to this trend was identified in the second time interval $[-0.3, 0]$ s, during which the three main features of movement preparation were captured: synchronization in the theta and gamma bands, and desynchronization in the alpha and beta bands, as discussed in the ERDS analysis. In this interval, a greater number of significant connections was observed when the movement was performed with the left upper limb. To assess whether this pattern is consistent across different motor dominance profiles, it is proposed that the experiment be replicated in future studies with left-handed participants.

In this interval, it was also found that, across all frequency bands, connectivity was greater in the L→R direction than in the R→L direction. However, in the subsequent time interval $[0, 0.3]$ s, this pattern changed and became dependent on both the type of movement and the limb used to perform it. When lateral and vertical grasps were evaluated regardless of the executing limb, greater connectivity was observed in the L→R direction for the lateral grasp and in the R→L direction for the vertical grasp. When the executing limb was taken into account, greater L→R connectivity was found for movements performed with the right upper limb, and greater R→L connectivity for movements performed with the left upper limb. These findings may reflect a functional lateralization of preparatory motor control, possibly linked to dominant-hemisphere specialization and interhemispheric coordination mechanisms.

Intra- and interhemispheric connections across frequency bands did not appear to follow a specific pattern associated with the type of movement performed or the limb used. In contrast, a more coherent organization of connectivity was observed with respect to the temporal position of the evaluated interval. This temporal modulation is consistent with the dynamics of feedforward and feedback control mechanisms described in the Introduction, which are particularly relevant in the context of self-initiated movements^{1–4}. When mapped to cortical regions, the observed patterns involved primarily the left and right primary motor cortices (approximated by C3 and C4) and midline structures such as the supplementary motor area (approximated by Cz), which are known to play a crucial role in the preparation and initiation of voluntary actions^{68–71}. Moreover, reach-and-grasp movements are supported by parietofrontal networks, including the dorsal premotor and posterior parietal cortices, which contribute to the selection of motor plans and the integration of sensory information^{72,73}. The motor task evaluated in this study allowed participants to freely decide when to initiate the movement and which upper limb to use, thereby engaging a complex decision-making process. It is suggested that this volitional control contributed to the temporally structured patterns of effective connectivity observed across conditions.

To investigate whether specific patterns underlie the preparation and execution of lateral and vertical reach-to-grasp movements, connectivity profiles were compared between the two movement types, and connections consistently observed during movements executed with the right or left hand were identified. Table 4 summarizes these effective connections shared between the limbs for each type of movement. Vertical grasping was observed to generate a higher number of shared connections in all frequency bands during the initial analysis window $[-0.6, 0.3]$ s compared to the subsequent preparatory phase $[-0.3, 0]$ s. In contrast, lateral grasping exhibited the opposite pattern, with a greater number of shared connections in the $[-0.3, 0]$ s interval.

This temporal divergence may indicate that vertical grasping requires an earlier and potentially more prolonged preparatory process than lateral grasping. Such a difference could reflect distinct cognitive and sensorimotor demands associated with each type of movement, potentially involving earlier activation of planning-related networks in the case of vertical grasping. This interpretation aligns with the asymmetry previously observed between right- and left-hand executions during the late premovement interval $[-0.3, 0]$ s, and supports the

notion that the temporal distribution of effective connectivity is shaped by the internal processes required to voluntarily select, prepare, and initiate a specific grasping action^{68,69,74}.

In this part of the analysis, it was found that during the early movement execution phase ([0, 0.3] s), certain effective connections emerged exclusively for either lateral or vertical grasping. Notably, the central electrode Cz, often taken as an approximation of the medial premotor cortex and supplementary motor area (SMA), exhibited condition-specific directed connectivity: toward Cp1 (overlying left somatosensory/motor regions) during lateral grasping and toward Cp6 (overlying right somatosensory/motor regions) during vertical grasping. This pattern suggests that medial premotor and SMA activity may exert task-dependent influence on lateralized sensorimotor regions during the initiation of motor execution^{68–70}. Such integrative control by SMA and adjacent medial motor areas is consistent with their established role in initiating and coordinating self-paced voluntary actions.

In general, for both types of movement, the number of shared connections between limbs decreased in the final time window ([0.3, 0.6] s). Nevertheless, lateral grasping continued to show a higher number of consistent connections between limbs compared to vertical grasping. These results may reflect differences in the coordination demands of each grasping type during the early execution stage, consistent with the idea that task-specific motor strategies shape the temporal distribution of effective connectivity^{73,75}.

The analysis of the temporal evolution of effective connectivity, based on Laplacian-transformed EEG signals, revealed the presence of connections that remained stable across the three analyzed time step windows (premovement, movement onset, early execution, and execution). These stable connections appeared to be specific to both the type of movement and the limb involved in its execution. Notably, a greater number of persistent connections were identified during lateral grasping compared to vertical grasping. This pattern may reflect a more consistent cortical coordination strategy for lateral movements, whereas vertical grasping could require more dynamic reconfigurations of effective connectivity throughout the execution period.

Within the set of stable connections found for lateral grasping, those involving channel C4, corresponding approximately to the right primary motor cortex (M1), were particularly notable. Depending on the limb used, the direction of connectivity varied: C4 projected to F3 (approximating left dorsolateral prefrontal or premotor regions) when the movement was performed with the left upper limb, and to Fc1 (left premotor/sensorimotor area) when executed with the right. Another relevant channel was Cp1, overlying the left posterior parietal cortex (PPC), which exhibited statistically significant connections in all frequency bands during left-limb lateral grasping, and in all bands except gamma during vertical grasping. The direction of Cp1's connectivity also depended on the movement type: toward Cp6 (right PPC) in lateral grasping and F4 (right premotor area) in vertical grasping. These patterns suggest that regions corresponding to C4 and Cp1 play central roles in coordinating motor planning and execution, with their effective connectivity adapting dynamically depending on the movement type and the limb used^{72,73,75}.

To further explore the early dynamics of cortical connectivity, the transition from premovement to movement onset (i.e., the interval from -0.3 s to 0.3 s) was examined. Specifically, connections consistently observed across all five frequency bands during this temporal interval were identified. Most of these connections appeared to be specific to the motor condition and were not shared across different movement types or executing limbs. For the lateral grasp performed with the left upper limb, only the connection from C3 to F4, approximating left primary motor cortex (M1) to the right dorsolateral premotor cortex (PMC), fulfilled this criterion. In contrast, the lateral grasp performed with the right upper limb showed three consistent connections: C4 $\rightarrow$ F4 (right M1 $\rightarrow$ right PMC), C3 $\rightarrow$ Fc6 (left M1 $\rightarrow$ right premotor/sensorimotor), and C3 $\rightarrow$ Cz (left M1 $\rightarrow$ SMA/medial premotor cortex). For the vertical grasp executed with the left upper limb, five stable connections were identified, involving premotor and posterior parietal regions (Cp5 $\rightarrow$ F3, Fc5 $\rightarrow$ Cp5, Fc5 $\rightarrow$ Cp1, F4 $\rightarrow$ Cp6, and Fc5 $\rightarrow$ P3), and for the vertical grasp performed with the right upper limb, three connections were consistently present (F3 $\rightarrow$ F4, F3 $\rightarrow$ Fc1, and Fc2 $\rightarrow$ Fc1), with F3 $\rightarrow$ Fc1 observed in both right-hand tasks. These patterns suggest that the direction and stability of effective connectivity are highly task- and limb-specific, reflecting the dynamic recruitment of cortical motor and parietal networks to coordinate planning and execution of grasping action^{72,73,75}.

The evaluation of out-degree and in-degree patterns provides valuable insights into the spatial and temporal organization of effective cortical connectivity during both lateral and vertical grasping actions. Detecting nodes that predominantly act as sources of information (high out-degree) or as recipients (high in-degree) allows us to infer the functional contributions of particular cortical regions in the preparation and execution of upper-limb movements.

For lateral grasping, channel F3, located over the left lateral frontal cortex, stood out during left-hand trials, particularly in the premovement window ($t = [-0.6, 0]$), where it consistently appeared as a major origin of outgoing connections across all frequency ranges. This finding points to a possible top-down modulatory role of frontal cortical areas in preparing contralateral actions, consistent with the established participation of premotor and supplementary motor areas (SMA) in motor planning^{72,73,75}. At the same time, F3 also emerged as a significant hub for incoming connections (in-degree), underscoring its integrative role within the sensorimotor network during this preparatory period. Additionally, Cz became an important recipient immediately preceding movement onset ($t = [-0.3, 0]$), which may indicate the recruitment of midline motor areas as the execution phase approaches.

For right-hand lateral movements, no single channel displayed the strong dominance observed for F3 in left-hand trials. Instead, the connectivity profile appeared more distributed, with joint contributions from C3, C4, Fz, F3, and Cp1 during the premovement out-degree stage, followed by a shift toward C3 and C4 predominance during execution. Such a distributed organization may indicate interhemispheric coordination in right-hand actions or reflect distinct cortical control strategies between dominant and non-dominant limbs^{73,75}. In the in-degree analysis, Fc5 and Fc6 gained prominence in the preparatory phase but declined around movement

onset ($t = [0, 0.3]$), later regaining involvement during the late execution period. This evolving pattern may correspond to changing contributions of frontocentral areas in integrating sensorimotor information and updating motor output throughout the task.

In vertical grasp trials with the left hand, frontal (Fz, F4) and central (Fc5) channels consistently acted as key sources of directed connectivity across frequencies and time windows. This sustained engagement may reflect an increased contribution of frontal and premotor regions to the planning and initiation of vertical actions, possibly due to the additional postural adjustments demanded in the vertical plane⁷⁴. Regarding incoming connections, posterior channels such as P3 and Cp6 became prominent during execution ($t = [0, 0.6]$), which could indicate the participation of parietal areas in integrating sensory information to refine grasp control⁷³.

For right-hand vertical movements, channels C4 and F3 consistently showed elevated out-degree values across conditions, identifying them as principal initiators of cortical activity. Their recurrent participation across frequency bands and temporal windows underscores their potential importance in motor control, with C4 potentially reflecting contralateral coordination of the limb and F3 contributing to preparatory and planning-related processes^{74,75}. In the in-degree analysis, channels such as Fc5, Fc6, and Cp1 emerged as relevant targets at different stages, suggesting a dynamic redistribution of integrative roles within the cortical network. This flexible engagement appears to facilitate the transition from preparatory to execution-related control mechanisms⁷³.

Overall, the findings indicate a task- and phase-dependent organization of effective connectivity in upper-limb movements. Although certain cortical hubs (e.g., F3, C4, Fc5) repeatedly contribute to the network, their apparent roles as sources or recipients vary with time, hand dominance, and movement direction. The predominance of frontal activity during preparation, together with the recruitment of parietal regions during execution, is consistent with established sensorimotor hierarchies^{73–75} and underscores the value of Partial Directed Coherence (PDC) for examining directional neural interactions in motor behavior.

Future work should investigate how these connectivity dynamics are influenced by factors such as task complexity, motor learning, or neurological disorders, and whether targeted interventions on these cortical hubs could promote motor recovery in rehabilitation contexts.

Corticomuscular connectivity was subsequently examined to characterize the dynamic interaction between cortical activity and muscle activation throughout movement execution. While numerous previous studies on corticomuscular coupling have concentrated on discrete time points, typically around movement onset or steady-state execution^{23,24,76,77}, several investigations involving isometric motor tasks have reported weak but significant corticomuscular coupling even prior to movement onset, suggesting that descending cortical drive may already be functionally engaged during motor preparation^{41–44}. These observations indicate that CMC may not only reflect ongoing sensorimotor integration but also preparatory communication between cortical and peripheral structures. In the present study, the analysis was designed to capture the temporal evolution of this interaction. By continuously assessing connectivity over time, differences associated with grasp type (lateral vs. vertical) and the executing limb (left vs. right upper limb) were identified. This time-resolved approach provides novel insights into how neural control strategies may vary according to the motor context.

While corticomuscular coupling has been shown to involve both afferent and efferent pathways, the present analysis focused exclusively on efferent connectivity, which is associated with feedforward motor control²⁴. This methodological choice was motivated by the aim of the study, which is more closely related to the execution of voluntary movements than to the sensory feedback processes typically linked to afferent corticomuscular connectivity^{76,77}. Previous research has demonstrated that modulations in corticomuscular connectivity are task-dependent, reflecting variations in attention, precision, object compliance, displacement, force, and motor learning^{23,77,78}. In accordance with these findings, the present analysis examined the temporal evolution of PDC across different grasp types and compared its dynamics between movements executed with the left and right upper limbs.

The effective connectivity from the EEG to the EMG channels was found to decrease across all frequency bands from the premovement phase to the end of the execution phase, for both types of movement. Although visual inspection of the PDC curves suggested some differences, these did not clearly reflect the contralateral pattern expected from previous coherence^{76,79} and connectivity⁸⁰ analyses. Moreover, since these differences did not reach statistical significance, definitive conclusions regarding lateralization cannot be drawn.

To further characterize the temporal evolution of corticomuscular interactions, a complementary statistical analysis was conducted to identify the specific time points at which the effective connectivity between EEG and EMG significantly deviated from baseline levels. Non-parametric Wilcoxon tests were applied at each time point to compare the corticomuscular PDC values during movement execution against those recorded during the fixation period, with correction for multiple comparisons to control for Type I errors. Contrary to the initial expectation of observing increased corticomuscular connectivity around movement onset, the results revealed a consistent decrease in EEG-to-EMG effective connectivity across frequency bands once the movement began. This pattern suggests that, rather than reflecting a progressive recruitment of cortical output, the descending drive measured by PDC diminishes as movement execution proceeds.

Given that PDC quantifies the directed predictability between cortical and muscular signals within a multivariate framework, the observed reduction should not be interpreted as an absence of corticospinal communication, but rather as a relative decrease in the directional information flow from EEG to EMG once muscular activity becomes dominant. One plausible explanation is that, during execution, the EMG signals become increasingly self-predictive—meaning that their current activity can be more effectively predicted from their own recent past or from other muscles' activity than from cortical inputs. This phenomenon is consistent with the view that while cortical activity plays a key role in initiating movement, the subsequent maintenance and coordination of ongoing motor output rely more heavily on subcortical or spinal mechanisms that stabilize and sustain the action without continuous cortical supervision^{43,44,71}.

A possible explanation for the observed reduction in connectivity, as well as the absence of statistically robust lateralized effects, could involve not only the high complexity of the motor task but also methodological aspects inherent to using a directed, multivariate measure such as Partial Directed Coherence (PDC). Unlike traditional undirected coherence methods, which evaluate coupling between isolated signal pairs, PDC is based on a multivariate autoregressive (MVAR) model and estimates *partial* and *directed* influences among all signals included in the model^{28,81}. In a context involving multiple muscles and self-initiated, dynamic actions, the past activity of a given muscle or of other muscles may account for a larger proportion of variance in that muscle's EMG than cortical inputs, especially around movement onset when multiple feedback loops and intermuscular interactions are active^{36,82}. Consequently, the EEG→EMG PDC may appear reduced relative to undirected coherence even when cortical contributions are present. This methodological consideration likely interacts with task complexity, which in our study required precise control and multijoint coordination, including voluntary selection of both movement onset and limb use, potentially shaping the temporal and spatial profile of corticomuscular connectivity observed throughout task execution.

Overall, these findings highlight that the absence of statistically significant premovement effects and the reduction in directed connectivity following movement onset should be interpreted in light of both physiological and methodological factors. The present results do not exclude the possibility of preparatory corticomuscular interactions but suggest that such interactions may be more subtle or distributed across multiple neural and peripheral sources, making them less detectable through directed measures like PDC. Future studies integrating complementary metrics—such as coherence, Granger causality, or information-theoretic approaches—could help determine whether similar temporal trends emerge across methods, thereby refining our understanding of how descending motor control evolves from planning to execution.

In summary, this research investigated the temporal evolution of effective intracortical and corticomuscular connectivity during self-initiated reach-and-grasp movements by combining ERDS analysis and Partial Directed Coherence (PDC) applied to surface Laplacian-transformed EEG and rectified EMG signals. The experimental paradigm engaged internal decision-making processes by allowing participants to freely choose when and with which limb to execute the motor task. This design enabled the assessment of connectivity patterns related not only to motor execution, but also to movement planning and intention. The results revealed frequency- and time-specific connectivity profiles, with distinct patterns across movement types, execution side, and analysis windows. Notably, increased theta and gamma synchronization, along with alpha and beta desynchronization, were observed during the premovement interval. These spectral changes coincided with an increase in effective connectivity, particularly in the L→R and L→L directions during the late premovement window. Furthermore, task-specific and stable connections were identified across different temporal windows, suggesting that both dynamic reconfigurations and persistent pathways contribute to voluntary motor control. These findings underscore the importance of evaluating directional interactions across multiple levels, from local spectral dynamics to broader corticomuscular integration, and offer new insights into the temporally structured nature of neural processes underlying self-initiated motor actions.

Future efforts will focus, first, on replicating the experimental protocol with left-handed participants to assess whether limb preference truly influences the connectivity patterns hypothesized in this discussion. Second, the methodological strategy presented in the current study will be applied to restrict the focus to corticomuscular connectivity, using only the 16 electrodes included in the autoregressive model estimation, to deepen the analysis of the observed relationships between connectivity and limb laterality. Third, future investigations will explore how connectivity varies during tasks involving inhibitory control, decision-making under time pressure, or responses to external stimuli. These paradigms are expected to help disentangle internally generated motor control from stimulus-driven regulation. Fourth, we plan to acquire signals using a protocol in which participants are externally guided regarding movement onset, allowing a direct comparison of the dynamic connectivity patterns obtained with Partial Directed Coherence under both protocols (self-initiated vs. externally guided). Lastly, the development of individualized estimates of functional connectivity could be pursued in future work, particularly to inform clinical applications or personalized brain computer interfaces designed to predict or support intentional motor actions.

Methods

This study employed an experimental protocol designed to record EEG and EMG signals from 20 healthy right-handed participants during self-initiated reach-and-grasp movements, in which participants autonomously decided which upper limb to use. The protocol enabled the motor task to be performed under conditions that approximate natural movement execution, with movement initiated by the participant's intrinsic intention.

The acquired signals were subsequently preprocessed to reduce noise and ensure data quality. This process included the removal of artifactual or noisy trials, as well as the exclusion of trials that did not meet the experimental criteria for inclusion. Following this, the EEG signals were transformed using the surface Laplacian, which enhances high spatial frequency components (i.e., activity localized to a small cluster of electrodes) and attenuates low spatial frequency components (i.e., widespread activity across most or all electrodes). This transformation improved the suitability of the dataset for investigating interelectrode connectivity^{56–58}.

Movement onset was identified using signal features derived from EMG power changes and the variance captured by the Common Spatial Patterns (CSP) method. This information enabled the development of event-related desynchronization/synchronization (ERDS) analyses of transformed EEG signals, time-locked and centered on movement onset, as well as partial directed coherence (PDC) analyses to assess both intracortical and corticomuscular effective connectivity. The complete analysis pipeline followed in this study is illustrated in Figure 8.

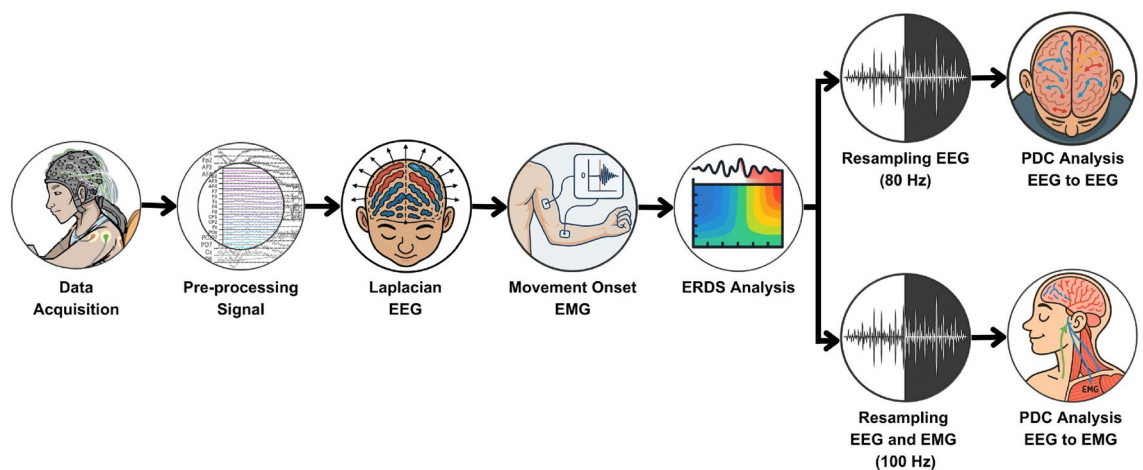


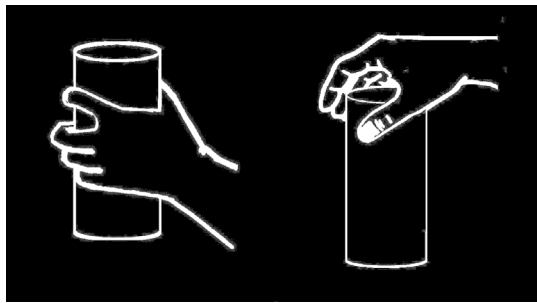
Fig. 8. Overview of the signal processing pipeline. EEG signals were acquired using the international 10/20 system, and EMG signals were recorded from six muscles involved in reach-and-grasp movements (three on each upper limb: the Lower Trapezius, Anterior Deltoid, and Pronator Teres) using the g.HIamp system (g.Tec, Austria). EEG data were preprocessed to remove noise and transformed using the surface Laplacian to attenuate low spatial frequency components. Movement onset was detected based on changes in EMG power. Transformed EEG and EMG signals were then segmented and time-locked to movement onset for ERDS and PDC analyses. This processing pipeline enabled the assessment of both intracortical and corticomuscular effective connectivity.

Experimental protocol

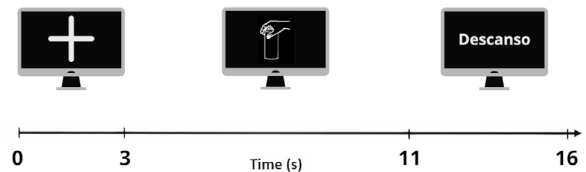
The signals analyzed in this study were acquired during the execution of a motor task in which participants decided when to initiate, and with which upper limb to perform, a grasp movement, in a vertical or lateral direction as instructed. During the experimental task, participants were seated in a comfortable chair in front of a computer screen that guided them through the various phases of the experimental protocol. Two rectangular foam pads, embedded with infrared sensors, were placed in front of the participants to support their forearms, thereby ensuring that all movements started from the same initial position and allowing the detection of movement onset. The target of the grasping movement was a tubular sensor device located between the foam pads and the computer screen, directly in front of the participant. This device was used to label each trial according to the type of grasp performed, using three integrated infrared sensors: one on the top, which, when activated, labeled the signals as a “vertical grasp,” and two sensors on the sides, which detected lateral grasps performed with either the right or left hand. Although the device could detect whether the lateral movement was executed with the right or left hand, all such trials were uniformly labeled as “lateral grasp”. The experimental setup is illustrated in Figure 10a.

Once the electrodes for signal acquisition were in place, participants were asked to sit comfortably in the anatomical resting position and to adjust the foam pads if necessary. They were instructed to follow the on-screen instructions throughout the entire experimental protocol. Figure 10b illustrates a participant performing a lateral grasp during the execution phase. In the image, the participant is shown using the left upper limb to perform the grasp, targeting the tubular sensor device as the goal of the movement. Figures 10c and 10d depict a participant in the anatomical resting position during the rest phase. In Figure 10c, the participant is viewed from the front, attending to the screen that provides instructions during the signal acquisition session. In Figure 10d, the participant is shown from the back, facing the screen with the upper back visible to reveal the EMG electrodes placed on the back. The illustration was created solely for explanatory purposes, and at no point were participants required to remove any clothing during the experiment.

The experimental protocol allowed for the acquisition of 300 trials per participant, divided into six series of 50 trials each. Each trial lasted 16 seconds and consisted of three phases. The first phase, lasting 3 seconds, was the attention phase, during which a fixation cross was shown on the screen. Participants were instructed to fixate on the screen and to avoid blinking or making abrupt movements to prevent unnecessary noise in the signals. The second phase, the execution phase, lasted 8 seconds, during which one of the images shown in Figure 9a was presented to indicate the type of movement to be performed. Participants were asked to wait approximately 3 seconds before initiating the movement, avoiding any strategy such as counting to estimate the time. The third and final phase, the rest phase, lasted 5 seconds and allowed participants to blink and adjust their posture for comfort before the next trial began. The timeline of each trial is illustrated in Figure 9b. After each block of 50 trials, participants were given time to rest as needed before continuing with the next block until all six blocks were completed. To ensure a standardized understanding of the task, a short familiarization phase was conducted before the experiment, during which participants completed up to five practice trials. This phase was stopped once the participant confirmed understanding of the task, so not all participants completed the full five trials. This familiarization was especially intended to address potential ambiguity regarding the instruction to wait 3 seconds before initiating the movement during the execution phase.



(a) Visual instructions specifying the required grasp type (either lateral or vertical) were displayed on the screen at the beginning of each trial. The corresponding symbol remained visible throughout the execution phase to indicate the intended action.



(b) A schematic representation of a single trial lasting 16 seconds. Each trial included an attention phase with a fixation cross, an execution phase during which the grasp cue was shown, and a rest phase indicated by the word “descanso” (“rest”). structure.

Fig. 9. Task instructions and trial timeline. **(A)** Visual cues indicating the type of grasp (lateral or vertical) were presented on a screen placed in front of the participant. **(B)** Timeline of a single trial lasting 16 seconds. Each trial consisted of three phases: an attention period, in which a fixation cross was shown; an execution phase, during which the grasp cue was displayed and participants voluntarily decided when to perform the reach-and-grasp movement and which hand to use; and a rest period, during which the word “descanso” (“rest”) appeared on the screen. Timing and phase structure were kept consistent across trials.

Task instructions

Participants were verbally instructed to maintain their gaze on the fixation cross during the attention phase and to avoid blinking or abrupt movements to minimize noise in the recorded signals. During the execution phase, they were explicitly told to wait approximately 3 s after the presentation of the movement cue before initiating the instructed reach-and-grasp movement, without using timing strategies such as counting. They were also instructed to use the 5 s rest phase to blink and adjust their posture for comfort before the next trial began. After each block of 50 trials, participants were encouraged to take additional rest as needed.

To ensure a standardized understanding of these instructions, a short familiarization phase was conducted before the experiment. During this phase, participants completed up to five practice trials, and the phase was concluded once the participant confirmed understanding of the task.

Data acquisition

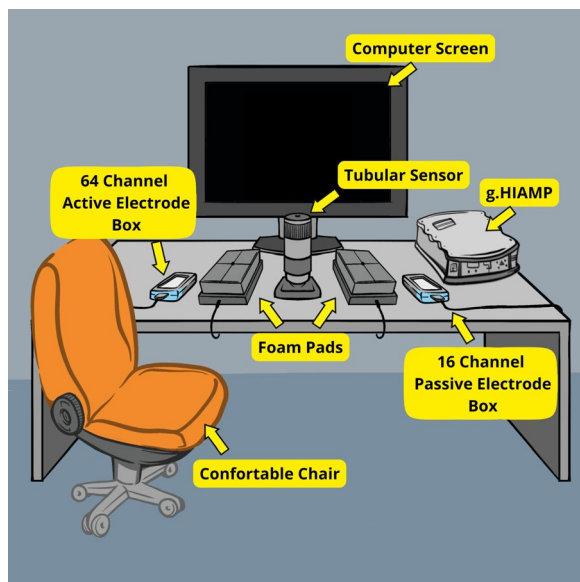
EEG data were recorded from 32 specific scalp locations, Fp1, Fp2, AF3, AF4, F7, F3, Fz, F4, F8, Fc5, Fc1, Fc2, Fc6, T7, C3, Cz, C4, T8, Cp5, Cp1, Cp2, Cp6, P7, P3, Pz, P4, P8, Po3, Po4, O1, Oz, and O2, according to the international 10–20 system. The ground electrode was located at AFz, and reference electrodes were placed on both earlobes. Surface electromyographic (EMG) signals were acquired using 12 electrodes positioned over three muscles on each upper limb: the lower trapezius, anterior deltoid, and pronator teres. EMG data were collected using Kendall™ Medi-Trace® pediatric ECG electrodes (Covidien, Dublin, Ireland). Prior to electrode placement, the skin was cleaned with alcohol to reduce impedance. The electrodes were pre-gelled, self-adhesive, circular in shape, and approximately 24 mm in diameter. A bipolar differential configuration was used for signal acquisition. For the lower trapezius, electrodes were placed vertically (one above the other), aligned parallel to the spine in accordance with the direction of the muscle fibers. For the anterior deltoid and pronator teres, electrodes were positioned side by side along the direction of the fibers. The interelectrode distance was approximately 2 cm, following standard recommendations⁸³. The electrode placement was guided by anatomical landmarks and carried out by trained personnel.

EEG and EMG signals were acquired simultaneously using the g.HIamp neuroscience research system (g.tec, Schiedlberg, Austria). The sampling rate was 1200 Hz. Signals were filtered to remove power-line noise and low-pass filtered at 500 Hz. In addition, the digital output from the foam pads and tubular sensors was recorded to identify both the movement onset and the type of movement performed. This sensor incorporates CNY70 reflective units with infrared emitters and phototransistors, operating within a 0.45 mm detection range⁸⁴.

The placement of both EEG and EMG electrodes is illustrated in Figures 10c and 10d. EEG electrodes are shown as part of the cap configuration, while EMG electrode locations are specifically highlighted over the corresponding muscles. More generally, Figure 10 provides an overview of the full experimental setup, including all electrode positions.

Participants

Twenty healthy participants (11 male, 9 female; age range: 18–45 years; mean age: 25.39 ± 6.63 years) were included in the study. All participants were free from any history of neurodegenerative disorders, had no prior experience with EEG or BCI-related experiments, and held no academic affiliation with the research team. Before participation, each individual received comprehensive information regarding the objectives of the study and provided written informed consent, in accordance with the Declaration of Helsinki. The research protocol



(a) Overview of the setup showing the screen for task instructions, the foam pads, the tubular device as the movement target, the signal acquisition equipment, and the comfortable chair used by participants.



(b) Lateral view of a participant performing a lateral grasp with the left upper limb.



(c) Frontal view of a participant in the anatomical resting position during the rest phase.



(d) Posterior view of the resting position, illustrating the participant facing the screen. The illustration shows the back uncovered to display EMG electrodes.

Fig. 10. Diagram of the experimental setup and participant postures during the reach-and-grasp task. Panels B D illustrate key moments of the protocol: **(B)** execution of a lateral grasp with the left upper limb, **(C)** anatomical resting position during the rest phase (front view), and **(D)** the same resting posture viewed from behind. The back is shown uncovered in **(D)** solely for illustration purposes; no participant was asked to remove any clothing during the experiment.

was reviewed and approved by the Institutional Research Ethics Committee at the Instituto Tecnológico y de Estudios Superiores de Monterrey (approval code: CA-EIC-2406-03).

EEG recordings were monitored throughout the experiment to ensure data quality. If participants exhibited interruptions such as coughing, sneezing, or signs of drowsiness, the recording was paused until the participant was able to resume the task in appropriate conditions. In such cases, the affected data segments were excluded from the analysis. However, if the interruption was severe (such as sustained drowsiness or an uncontrollable coughing episode), the experimental protocol was terminated for that participant, and all data acquired from them were excluded from the study.

Data preparation and pre-processing

The acquired signals were processed to reduce noise and artifacts that could compromise the analysis outcomes proposed in this study. Signal preprocessing was performed to segment the data into analysis windows and to reject noisy trials and channels. Initially, the signals were segmented into epochs corresponding to the duration of each trial (i.e., 16 seconds). The analysis window was defined to minimize the impact of artifacts caused by eye blinks and residual movements typically occurring at the beginning of the attention phase. These events are likely because, in continuous trials, the attention phase immediately follows the resting phase. While the image indicating the onset of the attention phase is being processed by participants, such movements may still be executed. To account for this, the analysis window was defined as [1, 11] seconds.

For the EEG signals, an artifact rejection procedure was developed and implemented by evaluating each channel across all trials for each participant. A channel was marked as noisy if the peak-to-peak amplitude exceeded 100 μV or if the standard deviation surpassed 50 μV ⁵⁸. The EMG signals were visually inspected on a trial-by-trial basis to ensure that no distorted signals were included in the analysis. Such distortions could arise from electrode placement issues, which may occur due to the nature of the motor task under evaluation or due to movements performed by participants during the resting phase. A visual inspection approach was preferred over automatic thresholding due to the highly variable morphology of EMG signals across subjects and motor conditions, which can hinder the reliability of fixed amplitude-based criteria.

Channels with more than 10% of trials marked as noisy were excluded. Additionally, participants with more than 30 noisy trials were discarded. This rejection procedure was applied to data from 23 participants, resulting in the exclusion of 3 participants due to an excessive number of discarded trials or channels.

To minimize ocular and other artifacts, we applied a combination of procedures: (i) task design and instructions, whereby participants were asked to maintain gaze fixation and avoid blinking during the attention and execution phases; (ii) automated rejection of noisy channels and trials based on signal characteristics (a channel was marked as noisy if the peak-to-peak amplitude exceeded 100 μV or if the standard deviation surpassed 50 μV ; participants with more than 10% noisy trials were excluded); and (iii) transformation of EEG signals using the surface Laplacian. Specifically, the spherical spline method described by Perrin et al.⁸⁵ was applied to compute the second spatial derivative of scalp potentials. This procedure enhances high-spatial-frequency components (i.e., activity localized to a small cluster of electrodes) and attenuates low-spatial-frequency components (i.e., widespread activity across most or all electrodes), thereby improving the suitability of the dataset for investigating interelectrode connectivity.

Movement onset with EMG

EMG signals were used to detect movement onset through a thresholding process followed by a classification step, which reduced the errors associated with thresholding alone. Given the high inter-subject variability in EMG amplitude, a fixed thresholding approach was insufficient for reliable detection across participants. To overcome this limitation, classification was employed to refine and strengthen the detection process.

For the thresholding step, the envelopes of the six differential EMG signals were first computed and then summed to obtain a single signal representing the overall muscular activity. The mean value of this summed signal was subtracted to remove the constant baseline offset. This normalization ensured that the signal fluctuated around zero, allowing for a consistent and unbiased thresholding procedure.

Movement onset was marked using the foam pads, which simultaneously served to support the participants' forearms in a resting position. A one-second window centered on this marker was then analyzed to determine the maximum amplitude of the signal. The envelope of the EMG signal was subsequently scanned backward in time to identify the first point at which the amplitude reached 10% of this maximum value. This threshold was empirically selected to accommodate inter-subject variability in EMG amplitude and allowed for a more accurate estimation of movement onset.

The classification process was implemented using a leave-one-out cross-validation strategy and a Support Vector Machine (SVM) classifier trained on Common Spatial Pattern (CSP) features extracted from the EMG signals. Features were computed using a 1-second time window across the frequency range of 2 to 350 Hz. Two class labels were defined for training: "movement", corresponding to the 1-second window following the onset detected by the thresholding process for each trial, and "no movement", defined as the last second before the presentation of the image that marked the beginning of the execution phase.

For each iteration, the classifier was trained on all trials except one, which was used for testing. The test trial was evaluated using a sliding window of 1 second with a step size of 0.01 seconds, starting from the onset of the execution phase (3 seconds after trial onset) and continuing until the end of the movement. At each step, CSP features were extracted and classified. The onset of movement was then defined as the first time point at which 50 consecutive windows were classified as "movement." This criterion provided a robust estimate of the true onset by filtering out isolated misclassifications. On average, this estimated onset preceded the foam pad marker by 240 ms (± 80 ms), reflecting the anticipatory nature of the EMG-based detection.

Time frequency analysis

A time frequency analysis was performed to examine task-related fluctuations in spectral power linked to neural activity. Based on the event-related desynchronization/synchronization (ERDS) framework⁸⁶, the analysis focused on identifying transient oscillatory dynamics associated with reach-and-grasp movements, with particular emphasis on the premovement phase, during which participants autonomously decided when to initiate the action and which limb to use. To increase spatial specificity and reduce the effects of volume conduction, the analysis was performed on EEG signals transformed using the surface Laplacian, which enhances local neural activity by emphasizing high spatial frequency components and attenuating widespread signals^{56–58}.

The ERDS analysis followed the methodology proposed by Marroquín et al. (2004)⁵⁹, which performs a time frequency decomposition of EEG signals and computes statistical deviations in logarithmic power relative to the average power during a defined baseline period. A significance index is then derived from these deviations to segment the time frequency plane into regions with uniform activation patterns.

Two segments were extracted from the Laplacian-transformed EEG signals. The baseline segment comprised the two seconds preceding the visual cue that marked the beginning of the movement execution phase (i.e., the [1, 3] s interval in Figure 9b). The movement-related segment also spanned two seconds and was centered on the actual onset of movement, including one second of premovement activity and one second of motor execution. The precise location of this window varied across trials and participants, depending on when each participant initiated the movement.

Time-frequency representations in the [4, 40] Hz range with 0.5 Hz resolution were computed for each electrode and trial within both signal groups using Morlet wavelets, defined as $\omega(f, t) = A \cdot m$ with $m = e^{-t^2/2\sigma^2} e^{2\pi i f t}$, $A = 2/\sum |m|^2$, and $\sigma = 0.2$. The Morlet wavelet was selected for its favorable balance between temporal and spectral resolution. Its Gaussian envelope in both domains makes it particularly suitable for detecting event-related desynchronization (ERD) and synchronization (ERS) patterns during motor task execution. Subsequently, the ERDS values relative to the reference interval were calculated independently for each electrode.

Partial Directed Coherence (PDC)

Partial Directed Coherence (PDC) is a method in the frequency domain designed to characterize the directionality of interactions among multivariate time series. It relies on the decomposition of partial coherences derived from multivariate autoregressive (MVAR) models. This approach is closely linked to the concept of Granger causality, which uses vector autoregressive modeling to capture temporal dependencies. In this framework, one signal is said to “Granger-cause” another if knowledge of its past improves the prediction of the second signal, indicating the presence of directed connectivity^{28,48,49,58}.

The implementation of PDC can be summarized in three main steps: the estimation of a multivariate autoregressive (MVAR) model of order ρ , the transformation of the model into the frequency domain, and the computation of the PDC values. To estimate the MVAR model, it is assumed that a set $S = \{x_m, 1 \leq m \leq M\}$ of M analyzed signals follows the MVAR representation. Specifically, for the multivariate time series $x(n) = [x_1(n), x_2(n), \dots, x_M(n)]^T$, where $n = 1, 2, \dots, T$ denotes time samples, the MVAR model of order ρ is applied, as shown in Equation 1⁸⁷.

$$x(n) = \sum_{l=1}^{\rho} A_l x(n-l) + v(n), \quad (1)$$

In this equation, A_l are $M \times M$ coefficient matrices, where each element $a_{ij}(l)$ quantifies the linear influence of the past value $x_j(n-l)$ on the current value $x_i(n)$. The term $v(n)$ represents a stationary multivariate innovations process with zero mean (without loss of generality). This modeling approach assumes that the analyzed signals are stationary over the analysis period^{48,58,88}. In practical applications, EEG signals are often assumed to be stationary within short temporal segments, typically spanning a few hundred milliseconds, which supports the suitability of this assumption for PDC analysis^{58,89,90}.

To further validate this assumption, we evaluated stationarity using Augmented Dickey Fuller and KPSS tests. Most channels satisfied both criteria; however, up to four channels per subject occasionally did not. Given the short 300 ms analysis windows, and in line with previous studies on movement-related EEG activity^{58,89,90}, we proceeded under the assumption of local stationarity. For each trial, a multivariate autoregressive (MVAR) model was then fitted to the data, with model order selected on a trial-by-trial basis using the Akaike Information Criterion (AIC). Model convergence was required in at least 70% of the trials per subject; subjects not meeting this threshold were excluded from further analysis. This criterion was satisfied for all EEG connectivity analyses, whereas in the corticomuscular analysis one subject did not meet this requirement and was therefore excluded.

Once the multivariate model has been fitted and its coefficients have been estimated, the frequency-domain representation of Granger causality is computed through Equation 2, where j denotes the imaginary unit (i.e., $\sqrt{-1}$), and I is the identity matrix of size $M \times M$.

$$\bar{A}(f) = I - \sum_{k=1}^{\rho} A_k e^{-j2\pi f k / f_s} \quad (2)$$

Finally, the Partial Directed Coherence (PDC) is computed using expression 3, which quantifies the directed influence from signal j to signal i at frequency f . This measure is obtained by normalizing the modulus of the frequency-domain coefficient $\bar{a}_{ij}(f)$ with respect to the total outflow from signal j , ensuring that the PDC represents a relative, frequency-specific connectivity pattern.

$$\pi_{ij}(f) = \frac{|\bar{a}_{ij}(f)|}{\sqrt{\sum_{m=1}^M |\bar{a}_{mj}(f)|^2}} \quad (3)$$

In this study, two PDC analyses were performed: one intracortical, using surface Laplacian transformations applied to 16 EEG channels (F3, Fz, F4, Fc5, Fc1, Fc2, Fc6, C3, Cz, C4, Cp5, Cp1, Cp2, Cp6, P3, and P4), and one

PDC analysis	Analyzed signals	Downsampled frequency	Temporal resolution
Intracortical	16 Laplacian-transformed EEG channels	80 Hz	Time windows: [−0.6, −0.3] s, [−0.3, 0] s, [0, 0.3] s and [0.3, 0.6] s
Corticomuscular	16 Laplacian-transformed EEG + 3 differential EMG channels	100 Hz	Continuous, 0.1 s step size

Table 8. Summary of the configurations used for Partial Directed Coherence (PDC) analyses. A fixed analysis window of 300 milliseconds was applied to both intracortical and corticomuscular signals.

corticomuscular analysis involving 19 signals, including the 16 transformed EEG channels and three differential EMG channels corresponding to muscles involved in reach-and-grasp movements of either the left or right limb. Both analyses were conducted across standard EEG frequency bands: Theta [4 8] Hz, Alpha (8 12] Hz, Low Beta (12 20] Hz, High Beta (20 30] Hz, and Gamma (>30 Hz).

For model fitting and the estimation of autoregressive coefficients, the Nuttall-Strand method was used. The model order ρ was determined using Akaike’s method, with the maximum order set to $\rho = 20^{88,91}$. Previous studies based on Granger causality have shown that reducing the sampling frequency decreases the required model order, thereby improving model fitting while preserving the timescale of the expected functional interactions (on the order of tens of milliseconds)^{32,33,58,92}. In this study, the sampling frequency was reduced to 80 Hz for the intracortical analysis and to 100 Hz for the corticomuscular analysis. These values satisfy the Nyquist criterion, as they correspond to twice the maximum frequency analyzed: 40 Hz for the intracortical connectivity and 50 Hz for the corticomuscular connectivity. Model fitting was performed separately for each trial, and convergence was evaluated by verifying that the multivariate innovations process corresponded to white noise, ensuring that the estimated MVAR models adequately captured the signal dynamics within each trial^{88,91}.

Regarding the duration of the analysis windows, various studies have employed temporal segments ranging from 50 to 100 milliseconds^{93,94} to 300 400 milliseconds^{33–35,67}, and even up to 1, 2, or 5 seconds^{31,37,92}. In the present study, 300 millisecond windows were used. This duration was chosen based on both methodological and empirical considerations. First, previous studies have shown that this duration is sufficient to capture task-related changes in brain activity during reach-and-grasp movements¹⁷. Second, segments of this duration can reasonably be assumed to meet the stationarity requirements of the MVAR model⁵⁸. Third, and importantly, among the different window lengths tested (e.g., 400 ms and 500 ms), the 300 ms window resulted in the highest number of trials with successful model convergence across all participants, making it a robust and practical choice for this study.

To evaluate the dynamic evolution of effective intracortical connectivity, PDC analysis was conducted across four discrete time intervals: [−0.6, −0.3] s, [−0.3, 0] s, [0, 0.3] s, and [0.3, 0.6] s. For each condition and time window, PDC was first computed at the single-trial level, resulting in a three-dimensional matrix of 16 × 16 channels by 40 frequencies per trial. These matrices were then averaged across trials to obtain a single group-level matrix representing effective connectivity across frequencies. When band-specific connectivity was analyzed, PDC values were further averaged across the frequency bins corresponding to each band. Statistical significance of the connections was assessed using the Wilcoxon signed-rank test, comparing each interval to a baseline window defined as the final 300 ms preceding the presentation of the visual cue, during which signals were recorded under the fixation condition (i.e., no movement was performed). For visualization purposes, only those connections whose mean PDC values exceeded 50% of the maximum significant mean PDC observed within each condition were displayed.

On the other hand, corticomuscular analysis was performed continuously from −1 to +1 s using a sliding window of 300 ms with a step size of 0.1 s. At each time point, PDC was calculated from surface Laplacian-transformed EEG signals (16 channels) to rectified EMG signals (3 channels corresponding to the evaluated limb), which resulted in a matrix of directed connectivity values 19 × 19 × 50, where the third dimension corresponds to 50 discrete frequencies sampled from 1 to 50 Hz. As in intracortical analysis, PDC values were averaged between trials and frequency-specific values were further averaged within predefined bands. The resulting group-level time-resolved estimates reflect the corticomuscular coupling throughout the time course. Each time point corresponds to the end of its analysis window; for example, the point at −1 s refers to the interval [−1.3, −1] s.

To examine differences in corticomuscular connectivity associated with the limb used for execution, the Wilcoxon signed-rank test was applied to compare the averaged connectivity values observed during movements performed with the right versus the left upper limb, separately for each movement type (lateral and vertical grasp). Unlike the intracortical analysis, no comparison was made against a baseline condition; instead, differences were assessed directly between the corticomuscular coupling patterns corresponding to each limb. To address the problem of multiple comparisons across all time points, a false discovery rate (FDR) correction was applied using the Benjamini-Hochberg procedure, with the significance threshold set at $\alpha = 0.05$. Only those time points that survived FDR correction were considered statistically significant. A summary of the signal types, downsampled frequencies, and temporal resolutions used in each analysis is provided in Table 8.

To determine whether the observed corticomuscular coupling from EEG to EMG was statistically significant relative to baseline, connectivity values were compared against a null distribution obtained from a 300 ms interval within the fixation cross period (2.6–2.9 s), that is, 100 ms before the end of fixation and prior to the visual cue indicating movement initiation. For each EEG→EMG pair, the mean PDC at each time point was compared to the corresponding mean PDC during the baseline window using the Wilcoxon signed-rank test (paired samples). To account for multiple comparisons across all pairs, p -values were adjusted using the

Benjamini–Hochberg false discovery rate (FDR) procedure with a significance level of $\alpha = 0.05$. Only time points that survived FDR correction were considered to represent statistically significant corticomuscular coupling beyond baseline levels.

Data availability

The dataset generated and analyzed during the current study is part of an ongoing research project and is not publicly available at this time. However, the data may be made available from the corresponding author upon reasonable request.

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Author contributions

J.A.S-B. collected the data, performed the data analysis, and prepared the manuscript. J.M.A. and O.M.-M. contributed to data interpretation and reviewed the manuscript. All authors approved the final version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to O.M.-M.

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